

Injuries in Routesetting:

A study of injury trends among routesetters in the indoor climbing industry.

by

Henry Bradford Dalby

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Dr. Nora Beltz _____
Assistant Professor of Department of Athletic Training

Dr. Sarah Fisher _____
Professor of Political Science and Chair of Political Science Department

Dr. Ed Davis _____
Professor and Chair of Geography and Earth & Environmental Science

Dr. Scott Boltwood _____
Professor of English and Chair of English Department
Director of College Honors Program

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CHAPTER I: REVIEW OF THE LITERATURE

1.1 Introduction

The indoor climbing industry has grown exponentially since the first indoor climbing gym opened in 1987.¹ In its latest growth spurt from 2013-2018 the industry grew by almost \$700 million dollars, approximately 7% per year, in the US alone.² A professional routesetting industry, composed of those individuals who regularly construct commercial climbing routes on indoor climbing walls, has emerged to meet the growing demands of the industry. As mainstream culture continues to popularize indoor rock climbing, the routesetting profession has grown to meet higher demands for quality commercial climbing experiences.³ The inclusion of indoor climbing in the 2021 Olympics demonstrates the international growth in the sport's popularity. With increasing coverage from media and visibility on the international Olympic stage, indoor climbing participation is predicted to continue increasing.⁴

The commercial success of indoor climbing facilities relies largely on their routesetting programs.⁵ All aspects of the climbing industry are professionalizing to capitalize on the booming market for the sport.⁶ With greater economic opportunities, large climbing gym corporations are emerging to meet consumer demand. El Cap network, for example, makes up one of the largest climbing gym chains in the United States with a total of 16 locations that include the gyms of Earth Treks, Movement, and Planet Granite.⁷

A key element in a climbing gym's financial success is its routesetting program which drives membership growth and customer retention.³ Thus,

demand for skilled routesetters is increasing and, in turn, driving the professionalization of the routesetting field. Ward Byrum, a routesetter of over 20 years and director of routesetting for Earth Treks Climbing and Fitness explains his perception on industry professionalization: “That theme of professionalization has helped us gain respect industry-wide [across the indoor climbing industry]. It’s enabled us to gain things like better pay and better benefits because of this specialization.”³ Increased demand for quality routesetters has elevated their compensation and career opportunities. In return, climbing gyms are demanding a better product output. However, informed regulation of routesetting is lagging behind and impeding industry growth. This is largely due to the lack of professional standardization in routesetting and, perhaps even more so, the lack of case-based knowledge (i.e., knowledge from routesetting-specific scenarios) about the profession.

Case based knowledge is the study of specific issues as they apply to routesetting, and it provides solutions to meet the unique situations found in the profession. Having been validated in previous applications, case based information can then effectively inform future discussions while intentionally accounting for the unique environmental context found in routesetting. Available literature fails to account for the unique context in routesetting when discussing workplace injury. Thus, better understanding of routesetting-specific labor is needed to support effective workplace standards and regulations and ensure that routesetters can continue to support the growing indoor climbing industry.

1.2 What is Routesetting?

“Routesetting,” at its very fundamental meaning as a verb, is the act of attaching modular climbing holds to an artificial climbing wall.⁸ As an emerging concept since the 1980s, routesetting has been called different things in different countries and at different times in history, including: route designer, builder, and course setter.⁵ The term “routesetter” has become the professional iteration of this concept and is considered the internationally recognized title for the occupation.⁵ While the scope of labor required of routesetters has evolved over time, this summary will identify existing industry norms to provide a generalized understanding of the profession as it currently exists. However, routesetting remains an expression of the surrounding culture. As such, routesetting will always be subject to the dynamic cultural fluctuations that will continue to shape new and evolving needs.

1.2.1 *The Role of the Modern Routesetter*

While indoor rock climbing was conceived as a training tool for the dedicated rock climbing athlete, its role has shifted over time. Modern commercial climbing facilities cater to a growing population of recreational climbers, as noted by routesetter and author Louis Anderson in his seminal work, *The Fundamentals of Routesetting*: “Today, the vast majority of new climbers experience climbing for the first time at their local artificial wall and many come to understand or identify themselves as climbers mostly, or even sometimes exclusively, through their experiences indoors.”⁵ Recreational climbers might visit

a climbing gym multiple times a week to workout, socialize, or take advantage of any of the other numerous amenities (e.g., restaurants, bars, cafes, etc.) modern commercial climbing gyms now provide.

Despite these peripheral amenities, rock climbing remains the main attraction for climbing gyms. To retain members and attract new ones, commercial climbing gyms regularly change the climbing holds out on the climbing walls (e.g., as frequently as every day and typically at least once every 1-2 weeks). However, rather than simply distributing holds at random on the wall, new routes are “set” to create unique challenges and stimulate their intended users. This is the primary role of the routesetter: to conceptualize, design, and construct new routes for climbers. Most climbing gyms offer new routes at least once per week for the customers, with larger gyms requiring more routesetting labor to maintain a consistent setting schedule. As larger facilities and corporate chains of climbing gyms become more common, it is increasingly possible for the professional routesetter to work full time in this capacity.

Understanding the scope of labor associated with routesetting first requires a basic anatomical understanding of a climbing wall (Figure 1). Modern climbing walls are built of concrete or wood and designed solely for the purpose of climbing. Commercial climbing walls are built with a grid of bolt holes, each with a t-nut installed on the back side. A t-nut (Figure 2) is a threaded sleeve designed to accept a certain size bolt, and modern climbing holds are attached to the wall using this bolt and t-nut system. Screws are often used, in addition to bolts, to fasten climbing holds to the wall. Modern climbing holds (Figure 3) are

made of polyurethane, polyester, fiberglass or wood--each with different properties and applications. The holds are designed to accept bolts and sit flush against a climbing wall when fastened using this t-nut or screw-on system. In this way, climbing holds can be attached to, turned on, moved on, and removed from a climbing wall--creating a variable three-dimensional environment on which to construct climbing routes.⁵

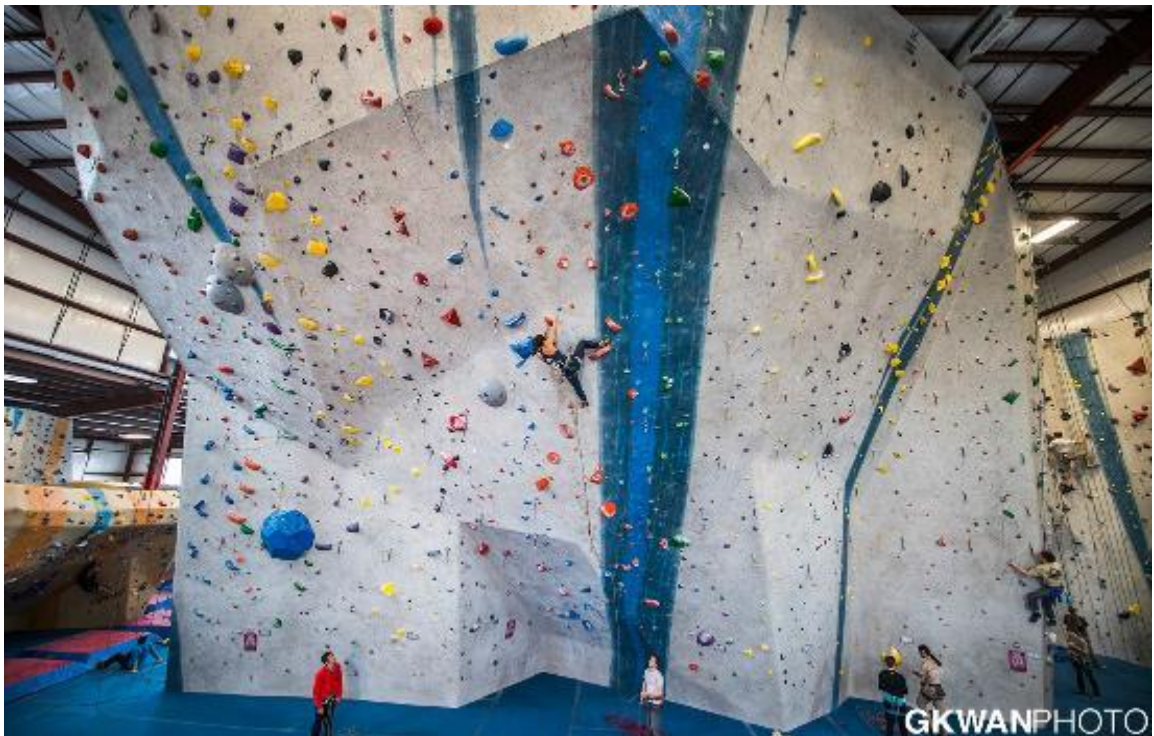


Figure 1: RockWerx climbing wall at Central Rock Gym, Watertown⁹



Figure 2a: Types of climbing wall t-nuts¹⁰

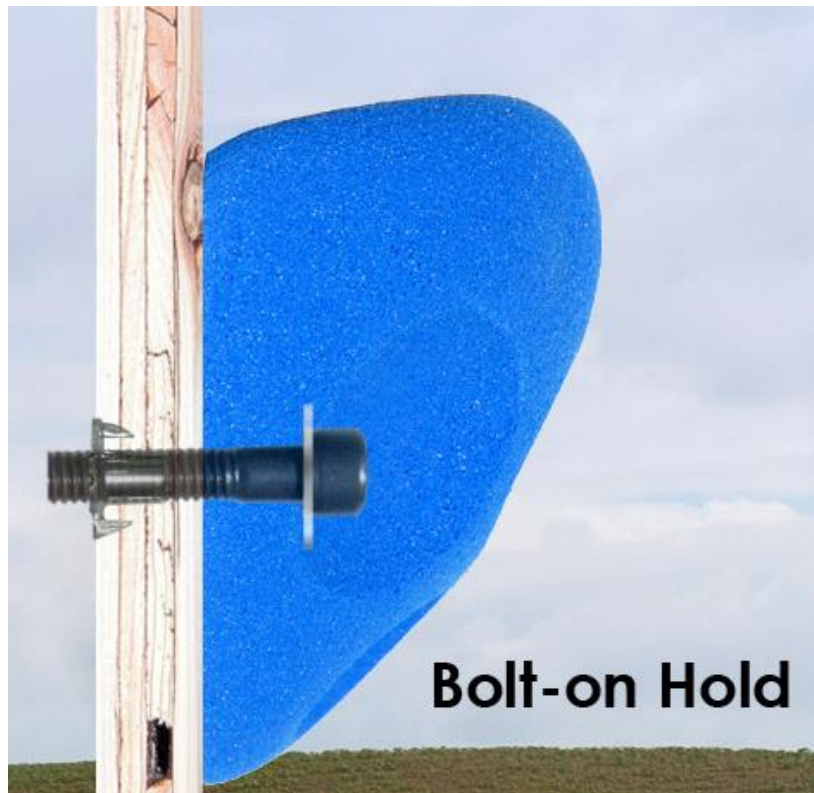


Figure 2b: Bolt on hold using bolt and t-nut system¹¹



Figure 3: Assorted climbing holds made by Rock Candy Holds¹²

1.2.2 Routesetting Tasks Explained

The labor associated with building a climbing route extends well beyond the act of attaching holds to the wall. Routesetters may be responsible for taking off old holds from the wall (“stripping”), washing holds, washing the wall, and organizing materials. All this occurs before even beginning to set a climbing route.¹³ Many routesetters perform a range of labor required throughout the process, from pre-setting tasks (i.e stripping, hold washing, and hold sorting) all the way through setting, forerunning, and cleaning up. Some gyms, however, hire dedicated employees to perform the less specialized labor (such as pre-setting tasks) so that routesetters might focus exclusively on their work of building quality routes. Commercial climbing gyms offer routes of different difficulty levels to cater to climbers of different abilities. Thus a routesetter must have diverse

climbing experience and a broad range of skills to meet the needs of diverse clientele.

Attaching holds is only the start of the process, and a significant portion of routesetting often occurs after this initial effort. While the first action of attaching climbing holds to the wall might be considered a “rough draft,” the final draft comes after an equally involved task of testing the route. Zeidelhack et al. found that most routesetters spend an average of 30-40% of the day on this process.¹⁴ Routesetters will typically climb the new routes to check for a variety of factors such as safety, difficulty, fairness, and even fun. During testing, routesetters may adjust existing holds or add new ones to meet their objectives for the route. This process of climbing and modifying the route is referred to as “testing” or “forerunning.” Changes made to a route during this process are often referred to as “tweaks.”⁵

1.2.3 Routesetting as Product Design

There are many tasks associated with routesetting, from the athletic elements of climbing, to the creative labor of designing a route, and the physical demands of constructing it. Thus, routesetters perform a variety of work as part of their job, described by former USA Climbing Chair of the Routesetting Committee, Chris Danielson, as “artistic expression, problem solving, and what is fundamentally product design.”⁵ Climbing routes are not just aesthetic but are also, and perhaps more importantly, functional. Routes are set with an intended purpose for an intended audience. Thus, there is both cognitive as well as

physical labor involved in setting a route. To implement a design, setting employs construction techniques to manifest the intention of the setter. Routesetting may involve using power tools and hand tools to attach holds to the wall, ascending and descending ladders and ropes, hauling heavy loads of climbing holds (both to the worksite and high onto the wall), climbing during forerunning, interacting with other team members to provide and receive criticism, and any other number of physical and intellectual tasks. Thus, routesetting requires a diverse set of highly skilled labor in addition to the creative demands of the professional.

Routes are the main product that climbing gyms offer to their users. Setting for a climbing gym can be compared to being a chef for a restaurant-- client satisfaction constantly demands new and stimulating products that reflect cultural trends. Each route contains new and unique qualities designed to challenge the climber. Although different setting environments (such as commercial versus competition setting) will present different objectives for a given routesetter, the scope of labor and medium for expression is fundamentally homogenous. For example, commercial routesetters may be more focused with providing a fun climbing experience while competition routesetters may be more interested in providing a show for the audience or a clearly divided field of competitors for scoring.⁸ Regardless of the different objectives for designing a route, both setters will be attaching holds to the wall to meet their goal.

Indoor rock climbing is more popular than ever, and the number of participants is only increasing. A greater number of routesetters are being employed to meet the needs of commercial climbers and increased volume of

recreational climbers. However, the profession is still in its development with variations in setting practices across the industry. The lucrative economy of indoor climbing places higher pressure on the routesetter to produce consistent quality products and operate according to industry best practices. Yet, without routesetting-specific, evidence-based recommendations, regulatory bodies are struggling to define best practices in routesetting, instead drawing from tangential bodies of work such as rock climbing and construction research.⁶ There remains a growing demand for research and case-based literature to inform the effective implementation and regulation of best practices in routesetting and to ensure the sustainable practice of profession routesetting as the industry grows.

1.3 The Professionalization of Routesetting

As the indoor climbing gym industry grows, industry leaders are allocating increased resources towards the professionalization of routesetting.¹ Furthermore, the increased economic activity of the industry is generating greater scrutiny towards the operational practices in climbing gyms.⁶ As industrial workers, routesetters are facing increased regulation enforcement by federal organizations such as the Occupational Safety and Health Administration (OSHA) in the United States.⁶ Without accurate industry data about the scope of routesetting labor, OSHA is enforcing regulations drawn from professions with similar practices, such as the rules about portable ladders, manlifts, and personal protective equipment.⁶ However, these regulations may not adequately address the specific needs of routesetting labor: what may be an efficient and safe way to minimize risk for carpenters working from ladders may not adequately facilitate

the productivity needs of routesetters. For this reason, industry regulatory bodies are seeking to better define best practices to both meet federal regulations while also facilitating professional routesetting to its highest quality.¹ Routesetting-specific evidence regarding its impacts on workers is needed in order to make evidence-based decisions regarding the functional needs of the profession while also meeting the highest standards for worker health and safety.

1.3.1 Existing Routesetting Literature

In 2018, Zeidelhack et al. conducted the first international survey studying routesetting as a profession: *Route Setters International*.¹⁴ The survey, published in the November 2019 edition of *Routesetter Magazine*, presents the most comprehensive information to date regarding the scope of labor associated with professional routesetting. In the study, Zeidelhack et al. identified elements of the profession that should be of critical focus for industry regulation.¹⁴ One of the article's most important conclusions is that the scope of labor involves diverse tasks that require skills for both construction work and rock climbing; routesetters are required to operate in the capacity of work-at-height professionals as well as climbing athletes within their daily work routine.³ While this conclusion might seem implied in the job title, Zeidelhack et al. provide substantial data for better understanding the specific implications of this skilled work. Specifically, the authors report that setters spend an average of 40% of the day forerunning when setting boulders, and 30% of the day forerunning when setting routes.¹⁴ Thus, the remaining 60-70% of time is spent performing other setting labor. The pilot study

conducted by the author of this paper revealed these tasks to include the following: ascending and descending ladders, lifting heavy equipment, operating power tools, ascending fixed lines, hauling equipment up fixed lines, and operating manlifts.¹⁵ Thus, professional routesetting labor is composed of near equal parts construction work, work-at-height labor, and work as a climbing athlete.

To better understand the extent of routesetting labor and its relationship with workplace injury, the author of the current study conducted a pilot survey in 2019. Respondents to the survey confirmed the findings of Zeidelhack et al. regarding the diverse scope of routesetting labor.^{14,15} Furthermore, over 70% of routesetters responding to the survey reported experiencing a work-related injury.¹⁵ These data not only demonstrate the prevalence of injuries in routesetting, but also suggest a high occurrence. High injury prevalence not only threatens day-to-day productivity, but also the long-term viability of professional routesetting.

Workplace safety measures may currently be inadequate for mitigating the risk of injury as only half of respondents to the *Routesetters International Survey* reported perceiving the level of safety in their work environment as high.¹⁴ If half of setters perceive high workplace safety yet over 70% of setters experience workplace injury, these “high” safety standards may be inadequate for protecting workers. Furthermore, of the participants in the current author’s pilot study, those setting less than five years demonstrated only a 58% rate of injury while those setting for five years or longer demonstrated over 85% rate of injury.¹⁵ This

stratification suggests that setters who have been working more years are more likely to experience injury. While this may be a reflection of elevated injury risk due to higher exposure accumulated over a longer period of time, it may also be the result of long-term wear-and-tear. Perhaps, also, the accumulated stresses from being a long-term rock climber compounded with the impacts of setting labor might result in overuse injury. Further, both Zeidelhack et al. and Dalby report that most setters are long-term climbers, with participants averaging 10-13 years of climbing experience between the two studies.^{14,15} The chronic stresses of routesetting represent the combined impacts of climbing and construction work, depicting a unique injury risk landscape for professional routesetters.

To understand the existing literature pertaining to routesetters, then, is to understand the work associated with both climbing and construction, as well as their respective relationships with injury. Literature will be examined within the following categories: general construction research, construction research that may specifically apply to routesetters, and general climbing injury literature. Between the intersection of these categories, the reader might gain a better understanding of the diverse impacts and concerns present in professional routesetting.

1.4 Construction Injury Literature

Routesetters attach holds to the wall using techniques adapted from traditional construction. One of the unique challenges to routesetting is performing this construction work in a vertical environment--using ladders and ropes to access walls of varying heights. Furthermore, the constant movement of

building materials introduces the additional element of moving heavy loads both to and from the work site in addition to their movement within the site (such as 20 meters up a rope to be installed at the top of a wall). These labor characteristics fall firmly within the US Bureau of Labor Statistics “Construction Laborers and Helpers” category, combining various skills associated with building, work-at-height, and general construction.¹⁶ With such a broad range of skilled labor, routesetters have been referred to as “Industrial Athletes” and even “Work-At-Height Professionals” to better capture the scope of labor involved with building climbing routes.^{3,17}

1.4.1 General Construction Injury Literature

While general construction literature does not fully address the entire scope of labor in routesetting, it should be applied to better understand those elements to which it does pertain. Laborers and general construction workers make up the largest group of injured workers in the United States, representing 29% of all workplace injuries.¹⁸ This statistic represents a dramatic overabundance of injury given that construction laborers only account for 6% of the active workforce.¹⁸ Furthermore, this study acknowledged that it had likely grossly underestimated the prevalence of injury among construction workers as a result of the federal injury tracking system it used for data.¹⁸ This federal system in the United States relies on self-reported injury statistics collected through hospital registration forms, and construction laborers are notorious for underreporting injuries.¹⁹ Up to 70% of occupational injuries are not captured by administrative databases, and this issue of underreporting only intensifies along

racial and ethnic lines: “Compared with white workers, black workers had significantly increased risk of injury when asked via self-report but not when using administrative data.”¹⁹ Thus, not only are construction workers at high risk of injury but the culture exhibits an epidemic of underreporting.

Causes of Construction Injury

There are three primary causes of injury to construction laborers: cutting or piercing against sharp objects, workers striking against or being struck by objects, and falling.¹⁸ Routesetters may be subject to all three of these leading causes of injury in daily labor: coming into frequent contact with sharp tools, operating in a vertical environment, and frequently moving materials through this vertical work site.⁵ Cutting or piercing against sharp objects represents the leading cause of construction laborer injuries and accounts for 26% of reported injuries.¹⁸ The cause of injury is most commonly “pieces of metal, razors, knives, power tools, and nails,”¹⁸ all of which routesetters might be exposed to on a regular basis.⁵ Furthermore, the fasteners used to secure climbing holds present a risk for injury as routesetters might be injured by sharp fasteners: either those that protrude behind the climbing wall or those that sit loose in the construction area. Author et al. report that 10% of sharp object injuries were from a nail or screw while over 12% of sharp object injuries were from power tools.¹⁸ With over 85% of routesetters using an impact driver, the workforce is regularly exposed to sharp object hazards.¹⁴

The second leading cause of injury among construction workers also poses distinct risk to professional routesetters: being struck by a falling object.¹⁸ Falling objects pose a risk of injuring workers as routesetters may frequently handle unsecured materials at height, the risk of dropping these objects presents a hazard to those operating below.¹⁸ Whether from 60 feet off the ground on a rope or from the top of a 15 foot ladder, a falling power tool or 30 pound climbing hold can cause significant damage to a worker on the ground if struck. Falling objects thus present a real hazard in the working environment of a routesetter.

Finally, routesetters are constantly exposed to the risk of falling: a mechanism that makes up the third leading cause of injury for construction workers.¹⁸ Two-thirds of routesetters report working from a rope most of the time, and all setting work includes the use of ladders.¹⁴ Working from a rope, when using appropriate safety systems and equipment, should present no risk of falling. Ladder workers, on the other hand, are typically untethered and rely solely on the ladder for support. Falling from ladders causes 27% of all fall related injuries to construction workers.¹⁸ Additionally, even workers operating on the ground face a high risk of injury from falling; trips or stumbles represent the second leading cause of fall injuries accounting for 19.8% of these injuries.¹⁸ Routesetters are also at risk of experiencing stumbling injuries since climbing gym flooring is made up of soft, padded foam. While this flooring helps to protect climbers against ground falls it also creates an unstable work terrain that may increase the likelihood of falling.¹⁷

All three of the leading causes of injury to construction laborers represent real hazards in routesetting, and even the lesser frequency mechanisms of injury represent common risks.¹⁸ While the fourth and fifth leading causes of injury among construction laborers represent a lesser percentage of total injuries (20% when combined), both overexertion and an object in the eye are regular hazards in the routesetting environment.¹⁸ The demands of installing heavy climbing holds from height can exert substantial strain on the shoulders, wrists, or elbows, posing a risk for overexertion. Zeidelhack et al. report that the leading two causes of injury in routesetters are “being tired” and “heavy loads,” accounting for 56% and 27% of injury, respectively.¹⁴ Furthermore, the use of impact drivers and other power tools creates the hazard of high-velocity debris that may come in contact with and injure the eye. Although 85% of routesetters report using impact drivers, just 35% use personal protective equipment (PPE), which includes eye protection.¹⁴ Unsurprisingly then, the third leading cause of injury among routesetters is missing PPE--aligning with the statistics reported by Welch et al about general construction labor.¹⁴ Thus, not only does the scope of routesetting labor fall within the parameters of construction work, but the injury trends in routesetting adhere to those found in the construction industry.^{14,18}

1.4.2 Construction Risk Mitigation Techniques

Quantifying construction labor injuries is productive in so much that it facilitates a better understanding for mitigating workplace risk. An engineering method, the CDIO-phase model (conceive, design, implement or construct, and

operate) aims to distill the phases of a construction project, and may be applicable to routesetting.²⁰ Based on this model, Soren Spangenberg concludes that there are two key opportunities increasing worker safety for any given project: 1) prioritizing worker safety during the design phase and 2) implementing cultural changes to the workplace during implementation and operation.²⁰ Decisions made before work begins on site result in up to 60% of all fatalities in construction work.²⁰ Thus, prioritizing worker safety when designing a given project can help mitigate the risk of injury.

By setting expectations for project timeline and worker conduct, injury risk can be reduced during the planning phase of a project.²⁰ Overworking construction laborers leads to injury.¹⁸ A study by Arlinghaus et al found that managing worker fatigue has a direct impact on reducing injury.²¹ Construction laborers who take rest breaks while working demonstrate a significantly longer time they are able to safely perform labor without incurring injury.²¹ Thus, a longer total rest break time results in a longer time on task without injury. By implementing a timeline and expectations for break time during the planning phase of a project, Spangenberg and Arlinghaus suggest that the risk of injury can be reduced.^{20,21} Routesetters are at a high risk of being overworked based on current industry practices in competition and commercial setting.¹⁷ As the indoor climbing industry pursues better practices for professional routesetting, it should consider implementing appropriate worker expectations and timelines to reduce risk of injury to routesetters.

Workplace culture also plays an important role in mitigating unnecessary risk. A clear example is the low frequency of PPE use among routesetters at just 35%.¹⁴ With the third leading cause of routesetting injury being missing or defective PPE, a shift of the industry culture to establish PPE use as the norm could help reduce this easily preventable hazard.¹⁴ In conjunction with industry changes to establish a standard of best-practices, creating cultural norms that promote safe operating practices can help improve the health and wellness of routesetters.²⁰

1.4.3 Construction Injury Literature Applicable to Routesetting

Routesetters are vulnerable to the same injuries as general construction laborers, and the epidemiology of construction injuries provides insights about general hazard trends. However, routesetting also involves highly specific, and often repetitive, labor.¹⁴ Thus, in addition to general construction workplace hazards, routesetters are vulnerable to specific mechanisms for injury. Two mechanisms, in particular, present high potential for injury: ladder falls and impact vibration. First, as work-at-height professionals, routesetters frequently use ladders to access the terrain, exposing individuals to a high risk of ladder-related injuries. OSHA has demonstrated a willingness to enforce safety legislation pertaining to proper ladder usage, and the importance of adhering to best practices is not only an issue of worker safety but of regulatory compliance.⁶ Second, routesetters are vulnerable to chronic injury caused by impact tool vibration, especially due to compromised postures while working at height.²² Over 85% of routesetters report using an impact driver for setting, and cultural

trends suggest this percentage of the population has been growing.¹⁴ Both ladders and impact tools present specific risks to routesetters.^{22,23}

Ladder Fall Injuries

Ladder injuries are a leading cause of injury among construction workers.²⁴ As of 2011, ladder fall injuries affect over 20,000 US workers each year.²³ Furthermore, over 81% of fall injuries requiring emergency medical care involve a ladder.²⁴ While ladders are used by many different professions, the construction industry has the highest rate of ladder related injuries.²⁴ Work-at-height professionals, such as routesetters, are particularly vulnerable to ladder injuries due to frequent exposure. More than $\frac{2}{3}$ of ladder fall injuries lead to lost workdays, and this creates not only an economic burden for the industry but a substantial public health burden.²³ On top of this, ladder fall injuries are largely preventable.²⁴

Ladders pose a distinct workplace hazard because of user behavior.²³ Environmental conditions rarely, if ever, increase this hazard; Lombardi et al report that in most cases of ladder fall injuries, the ladder was neither wet nor slippery and on a stable, level, and hard supporting base.²³ Routesetters who work indoors on unstable padded surfaces may be slightly more exposed to ladder fall injuries, but only marginally. Height also holds little sway over ladder fall injury prevalence, with the mean fall height among all cases being 7.5 feet above the ground.²³ In fact, the single factor most likely to increase risk of injury while on a ladder is activity. Workers who operate directly from a ladder, as

opposed to simply using the ladder for access to and from a site, are at the highest risk for ladder fall injury in the event of a fall.²³ Since routesetters are always working from ladders, their risk of injury increases.

Falls are typically caused by ladder movement, and activity both on and surrounding a ladder can cause instability and result in a ladder fall injury.²³ Most routesetting teams are composed of an average of 6.5 setters, all operating within a confined work site and often focused on separate tasks such as forerunning, setting, or moving equipment.¹⁴ This high level of physical movement within a confined space elevates the risk of causing a ladder fall injury. Thus, ladders present a uniquely high risk of injury to routesetters. Unstable flooring, working directly from a ladder, and high levels of activity within a confined space surrounding the ladder all contribute to this elevated risk.

Impact Vibration Injuries

Impact vibration presents another unique risk to routesetters. Since over 85% of routesetters use an impact driver for setting, workers are at risk of impact-related injuries such as carpal tunnel syndrome, Hand-Arm Vibration Syndrome (HAVS), and tendonitis.²⁵ Chronic exposure to hand-transmitted vibration through the use of impact tools can lead to the development of HAVS.²² Resulting in loss of tactile sensitivity and dexterity in the fingers and hand, HAVS can lead to lost work time and healthcare expenses in addition to strain on the economy if enough workers are affected. Furthermore, since most routesetters

are climbers, a chronic injury such as HAVS could affect the individual's ability to participate in rock climbing.^{14,15}

However, several key factors contribute to the likelihood of developing negative symptoms: grip force, posture, and temperature.²² Since routesetters typically work in temperature controlled indoor environments, they are not exposed to the low temperatures that induce HAVS symptoms. However, due to awkward positions while working-at-height, setters may be vulnerable to HAVS caused by poor posture and improper grip force. Exposure to a high frequency impact tools is linked to wrist injuries while lower frequency impact tools affect the shoulder and elbow.²⁵ The intensity of the exposure over time as well as the specific vibration frequency has been associated with different chronic tendonitis along this kinetic chain.²⁵ Routesetters operating handheld impact drivers may be exposed to different frequencies of vibration depending on the specific tool.²⁵ Routesetters may be vulnerable to HAVS due to vibration frequency, intensity of exposure, and poor posture.

1.4.4 Why Construction Injury Literature is Inadequate

Routesetters remain distinctly vulnerable to injury not only due to the construction-related stresses on the body, but due to the additional physical strains of forerunning and climbing, which may combine to increase the risk of injury. With so many distinct characteristics of routesetting injury, construction injury literature fails to capture the whole picture. Construction injury research functions to outline certain risks to the workforce, but routesetters remain

vulnerable also to climbing specific injuries. To understand the routesetting injury landscape requires a comprehensive picture of construction injuries, climbing injuries, and their intersection.

1.5 Rock Climbing Injury Literature

Routesetting and indoor rock climbing are inextricably intertwined, each depending on the other. Routesetters supply the product, indoor climbing routes, that is demanded by the consumer: climbers. To meet the need for a high-quality consumer product, forerunning is an integral component in routesetting and perhaps the most important step in the process of setting a route. To be an effective forerunner requires many skills, both to poses the technical ability to test a climb and also to communicate and implement tweaks. Forerunning may thus be the most highly skilled component of routesetting labor, requiring a more diverse array of experience and proficiency than its construction labor counterpart.

1.5.1 The Routesetter as a Climber

The average routesetter has been climbing for 6 years prior to beginning routesetting.¹⁴ These years of physical stress as an athlete intersect with the demands of professional routesetting to create a unique injury landscape. In addition to contending with the hazards presented by construction work, routesetters are also vulnerable to climbing injuries, both as a product of their

background in the sport and the additional climbing stresses placed on the body during forerunning and other setting activity. While the field of climbing sports medicine is young, as is the sport, increased participation has prompted the growth of case-based literature to better understand and treat climbing injuries.

1.5.2 Rock Climbing Mechanisms of Injury

Sports medicine has long recognized the specificity of climbing injuries, both with their unique mechanisms and highly targeted injury sites.²⁶ Over the past twenty years, medical research has begun to produce clinical knowledge that can support discussions about these unique climbing injuries.²⁶ A 2007 epidemiological study of climbing injuries found that 50% of climbers had experienced at least one injury over the preceding year.²⁷ With 4.6 million people in the United States participating in indoor climbing in 2015, that represents 2.3 million people who may have experienced a climbing injury.⁶

The most frequently injured sites for climbers are the finger, shoulder, and elbow.^{27,28} These sites represent about 80% of all chronic overuse injuries.²⁸ Furthermore, some of these leading sites of climbing injury are also the most likely to experience chronic injury from the construction labor in routesetting. Specifically, the shoulder and elbow are vulnerable to chronic injury caused by impact tool vibration.²⁵ These overlapping mechanisms of injury make routesetters a doubly vulnerable population: susceptible to chronic stress from both climbing and routesetting.

Understanding the relationship between experience level, climbing frequency, and injury is an important factor for understanding how climbing injuries intersect with professional routesetting. However, studies have been unable to agree on a definitive relationship between climbing frequency and injury. In 2018, Grønhaug et al. conducted the largest climbing injury study to date, finding the highest rates of chronic injury among experienced and elite climbers compared with recreational and international elite climbers.²⁸ This trend would support increasing rates of chronic injury corresponding to increased climbing ability except for the sudden drop in injury prevalence among the international elite climber group (those with the highest climbing ability). While this may be due to a study limitations caused by a very small number of participants in this group, the study suggests another possibility: “it may be that the international elite climbers as well as the most experienced climbers in the elite group have more knowledge of how to train and know their body well enough to train hard without getting injured.”²⁸ However, with the exception of the international elite climbers, the study may support corresponding findings by Jones et al linking a higher climbing frequency with cumulative trauma to the upper extremities and resulting chronic injuries.²⁷

While the specific relationship between climbing frequency and chronic injury remains unconfirmed, data unanimously supports the conclusion that climbers who participate in the sport with a high frequency experience high rates of chronic injury.²⁶⁻²⁸ Since routesetters typically spend 30-40% of a workday forerunning routes, working an average of 1-3 days per week, the population is at

risk of chronic climbing injuries.¹⁴ Furthermore, the types of injuries most frequently experienced in climbing overlap with those prevalent among impact tool operators.²⁵ Climbing injuries may thus impact a worker's ability to perform forerunning labor in addition to the routesetter's ability to perform other tasks associated with routesetting.

1.6 Summary of Literature Review

Routesetters are subject to many workplace risks as part of their daily labor. With a diverse scope of labor extending from forerunning, to regular ladder use, to construction from a rope, routesetters are engaging in activities that can cause injury. Since, injured workers cannot meet industry demand, it is imperative for the indoor climbing industry to prioritize the health and wellness of its routesetters to ensure full-time productivity that can support commercial climbing growth. As demand for quality commercial routesetting increases alongside the nearly \$700 million yearly industry growth (in the U.S.), experienced routesetters have become an invaluable component of any successful climbing gym.² With indoor climbing scheduled to be included in the 2021 Olympics, participation in the sport is only expected to increase.⁴

However, existing injury literature fails to adequately address the needs of routesetters. By examining literature on climbing injury and literature on construction injury it is clear that routesetters are at risk in their daily activities. However, case-based routesetting injury literature is needed to explore the specific vulnerabilities of the population given the unique scope of labor. Routesetters are a crucial component of a booming economic industry, both in

the U.S. and across the world. Yet, there is little to no research about the short or long-term injury risk to the population. To support the growth of the industry, the economy, and the workforce, routesetters require case-based injury literature to inform effective regulatory practices that can support the professionalization of the industry as well as the health and wellness of the individual.

1.7 Incomplete Glossary of Routesetting Terms

Boulder - Shorter climbs (typically around 15 feet tall) that are designed to be completed using pads on the ground to protect the climber during falls.

Box - Another term for **Volume**.

Climbing Gym - A facility that offers indoor climbing, typically as the main purpose of the business but often in addition to other amenities such as traditional gym equipment, fitness classes, and food/drink venues.

Climbing Hold - An artificial, three-dimensional element designed to be used as a modular element on artificial rock climbing walls to aid the climber and serve as a hand-hold for climbers. Typically composed of polyurethane, polyester, fiberglass, or wood.

Climbing Route - A path of holds up a climbing wall that a climber must follow to successfully reach the top of the wall. Indoor climbing routes are designed to mimic the concept of outdoor climbing. Climbing routes are often marked using holds of the same color (monochromatic) or tape strips.

Climbing Wall - An artificial climbing wall built and designed with the explicit purpose of providing modular rock climbs to users.

Difficulty - A measure of the challenge of a climbing route, relying on the subjective opinion of the routesetters and, typically, consensus from those who climb it. Difficulty is typically measured on a numeric scale with higher numbers indicating increased difficulty. There are many different climbing difficulty scales across the world, with the most common being the Fontainebleau climbing scale, the Verm bouldering scale, and the Yosemite Decimal System. Many other

scales exist as well, and they all rely on the subjective input from the first ascensionist and other climbers to grade a route.

Fairness - A term used in routesetting to discuss how difficult a route might be depending on different sized climbers. Typically, a commercial climbing route will attempt to cater to a wider range of climber sizes so the majority of participants can complete the route without it becoming overly difficult or overly easy.

Forerun - The process of climbing a newly designed route to test for predetermined objectives such as safety, fairness, or fun.

Impact - Short for "Impact Driver." A hand tool with an internal hammering mechanism to provide bursts of torque when met with resistance. Used to fasten holds to the wall.

Pre-Setting Tasks - Labor that must be conducted before routesetting can begin, such as stripping old holds from the wall, washing the dirty holds so they can be used again, and removing stuck holds/bolts from the wall to provide a blank canvas for the routesetters. These tasks vary between different routesetting programs.

Route – Can refer generally to a rock climb of any type. Can also refer specifically to climbs that are designed to be completed using a rope for protection from falling: these climbs tend to be taller in nature than their shorter counterparts (**Boulders**).

Routesetting - The process of constructing a climbing route on an artificial climbing wall. Typically involving attaching artificial climbing holds to the wall with a specific placement and intention to create a unique climbing route.

Set - The act of putting holds on a wall to create a climbing route.

Setter - An individual who engages in routesetting.

Strip - The act of removing climbing holds from a climbing wall.

T-Nut - The permanently installed, female counterpart to a bolt on a climbing wall. Distributed across a climbing wall surface to allow the repeated attachment and removal of climbing holds.

Testing - Synonymous with forerunning.

Tweak - The act of altering a climbing route, typically during forerunning. This may involve turning, moving, or adding a hold to the route to better meet the forerunning goals.

Volume - A particularly large climbing hold designed to have other, smaller, climbing holds attached to it. Volumes are often made of plywood or fiberglass to allow for their relatively large size without being too heavy to maneuver during routesetting.

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CHAPTER II: INJURIES IN ROUTESSETTING

2.1 Abstract

The purpose of this study is to explore the causes of injuries among routesetters. 178 participants responded to an online survey asking about their routesetting injury history. Data were analyzed to explore why routesetters are getting injured, if using protective eyewear affects the likelihood of injury, and if scope of labor affects injury occurrence. 64% of respondents reported experiencing an injury related to routesetting.

The leading sites of injury were finger, shoulder, back, wrist, elbow and eye. The leading causes of injury were overworking, too heavy loads, and forerunning. These injury trends represent the cumulative physical strain from performing both climbing labor and construction labor in combination with each other during routesetting. Existing injury literature regarding climbing injuries and construction injuries supports these findings.^{1,2}

Additionally, eye injury data confirmed that these injuries are highly preventable.¹ Over 80% of respondents who experienced an eye injury were not using protective eyewear during the incident. Of those who reported an eye injury, 85% reported being more likely to use protective eyewear in the future indicating that the majority of respondents perceive protective eye-wear as an effective injury mitigation technique.

Finally, data on scope of labor suggested three major findings. First, higher setting frequency correlates to higher injury occurrence: those who set 3-5+ days per week reported the highest rate of injury (85-86%). Second, those

who specialize in boulder setting or route setting reported a lower rate of injury occurrence compared to those who do both. Third, those who set only commercially or spent equal amounts of time setting commercially and for competition reported a low injury occurrence (35% and 17%, respectively), compared to those who set mostly commercially and sometimes for competition (68-81%).

The results of this study identify clear trends of injuries in routesetting. Findings are consistent with existing injury literature and routesetting injuries mirror trends found in both rock climbing^{2,3} and construction labor.¹ The study identifies clear risk areas for routesetters that can be addressed to help improve the health and wellness of the individual, create a more sustainable profession, and increase industry productivity.

2.2 Introduction

The indoor climbing industry has grown exponentially since the first indoor climbing gym opened in 1987.⁴ In its latest growth spurt from 2013-2018, the industry grew by almost \$700 million dollars, approximately 7% per year, in the United States alone.⁵ Climbing's inclusion in the 2021 Olympics is only expected to increase participation in the sport, with increased media coverage and visibility on an international stage.⁶ The growing economy created by indoor climbing has generated many full-time climbing industry jobs.⁷ In particular, the demand for commercial climbing is generating a corresponding need for quality commercial routesetting.⁸

With higher demand for quality indoor climbing experiences, routesetters have elevated their value to the industry. However, this elevated value has come with an increased emphasis on professionalism in the workplace to ensure consistent product output.⁸ Routesetters can now generate a lucrative career with their work when, just several years ago, the industry may not have been economically successful enough to sustain full-time routesetting as a widespread and viable job. As routesetting becomes more widespread, there is an increased need for standardized best practices to ensure the health and wellness of the individual and to support the continued growth of the indoor climbing industry.

Routesetting is the job of designing and building artificial climbing routes. There are two primary subgenres of climbing route: “boulders” (short climbs to be completed with padded flooring to protect against falls) and “routes” (longer climbs to be completed using rope systems to protect against falling). Increased participation in both styles of indoor climbing has resulted in higher demand for quality routesetting. With the elevated value placed on routesetters and better compensation has come an increased demand for consistent product quality and output.

However, industry-wide best practices for routesetting are lagging behind its growth. The varying consumer needs in different climbing gyms have led to varying standards for routesetter health and safety standards on both an international and national level. Furthermore, the growing economy of indoor climbing has drawn increased scrutiny from regulatory bodies making this not only an issue of worker safety but of regulatory compliance.⁷

These existing programs have drawn from professions with similar labor characteristics to routesetting, such as construction, to inform workplace standards.⁷ However, such regulations may be inadequate for meeting the specific needs of routesetting labor and productivity which combines aspects of work-at-height professions, construction labor, and rock climbing to generate a unique injury risk landscape. More case-based routesetting literature is needed to generate effective regulations and industry best practices. Ensuring the long-term health of routesetters and the sustainability of the profession is imperative for supporting the prosperity of the indoor climbing industry.

In 2019, Vertical Life Magazine published the first study on routesetting labor in Routesetter Magazine.⁹ The study findings confirmed unstandardized labor characteristics across the international routesetting industry, including varying scope of labor and use of personal protective equipment (PPE). Significantly, the study identified a prevalence of occupational injury among routesetters.⁹ These findings confirm results from a pilot study conducted by the author of the current study in which over 70% of routesetters reported experiencing an occupational injury.¹⁰ As the indoor climbing industry professionalizes to meet the needs of a growing economy and also to adhere to health and safety standards, case-based literature is essential for developing appropriate programs for routesetter wellness.

We conducted the study with three purposes for understanding the relationship between routesetting and injury. The primary purpose of this study was to evaluate the prevalence of injury and explore potential cause(s) of injury

in routesetters. Additionally, we sought to examine the relationship between personal protective equipment (PPE) and routesetter injury. Within this topic, we specifically examined PPE use regarding eyes based on our analysis for type of injury in which eye injuries represented a significant portion of reported injuries while ear and head injuries did not. Lastly, we sought to examine how scope of labor may be related to worker injury.

Based on the work by Zeidelhack et al.,⁹ our own pilot study,¹⁰ and existing injury literature in both rock climbing² and construction,¹ we generated several hypotheses. First, we hypothesized that there would be work-related injuries in routesetters and that a leading cause of these injuries would be overworking resulting from the high physical demands placed on the body from the combined load of both routesetting and climbing. Second, we hypothesized that missing or inadequate PPE would be associated with an increase in injury occurrence. Finally, and third, we hypothesized that higher work volume would correspond to higher rate of injury.

2.3 Methods

2.3.1 Participants

178 participants responded to the survey. Study participants were routesetters over 18 years old. The study was approved by the Emory & Henry College Institutional Review Board under exempt status for anonymous data collection. The Institutional Review Board Approved Study Application can be found in Appendix A.

2.3.2 Survey Distribution

An anonymous web-based survey was distributed via social media and email and remained open from January 7th, 2020 until February 9th, 2020. On social media, the survey link was initially posted to the Facebook group “Routesetters Anonymous” and reposted four times while the survey remained open. The survey was also distributed to several prominent routesetting organizations involved in the professional development, regulation, and promotion of the indoor climbing industry. All participants were invited to share the survey with others.

2.3.3 Survey Development

This survey was developed in Qualtrics Core Experience Manager using questions from a spring 2019 pilot study, which were refined for use in this survey. Additionally, research conducted by Zeidelhack et al. in 2019 helped identify international trends as well as areas of research that might be of particular value to the industry and to the author’s research goals. Lastly, the final survey was reviewed by an expert in each of the following areas: sport-related injury, survey research methods, and routesetting.

The current survey asked participants to report their personal experiences with injuries in routesetting. The survey included multiple choice, select all that apply, fill in the blank, and open-ended questions in three categories: (1) demographic questions, (2) scope of labor questions, and (3) injury history

questions. A comprehensive list of the survey questions can be found in Appendix B.

Demographic Questions

Demographic questions aimed to define our study sample and included questions regarding age, ethnicity, gender, years climbing, education, and physical activity level. Physical activity level questions aimed to understand a participant's activity level outside of work. These questions included number of hours per week spent rock climbing outside of routesetting activities, number of complete days off from physical activity per week, and activity level outside of rock climbing/routesetting, categorized as sedentary (less than 30 minutes of exercise per day), moderate (30-60 minutes of exercise per day), and high (over 60 minutes of exercise per day).

Scope of Labor Questions

Scope of labor questions aimed to identify a participant's typical work at the time of the survey and also to depict their workplace environment. These questions fell into two categories: workplace questions and specific labor questions. General workplace questions included the following: years routesetting, number of gyms worked for, type of climbing gym, location of primary gym, employment type, and compensation. Individual scope of labor questions included days per week setting, percentage of time spent setting routes versus boulders, percentage of time spent setting commercially versus for competition, number of routes/boulders per day, height of route/bouldering walls,

activities during work, and tool use. For activities during work, respondents were asked to check all that apply from a provided list and were also invited to write-in answers. Those who reported tool use were asked to identify the type of tool used for bolts.

Injury History Questions

Respondents were asked questions about healthcare (i.e., access to health insurance, access to subsidized health care, and healthcare seeking behaviors), daily health and safety routine, and injury experiences. These questions aimed to identify specific behaviors and environmental characteristics that might correlate to workplace injury. This section asked questions about PPE use (i.e., eye protection, ear protection, and helmet), warm-up before labor, perceived level of workplace safety, perceived industry shifts regarding routesetter wellness, and changing workplace standards about PPE use.

Injury experience questions aimed to identify a participant's routesetting injury history. These questions asked if the respondent had ever experienced a routesetting injury, estimated number of injuries over the past year, estimated number of career injuries, and location of the(se) injury(s) (i.e., back, shoulder, finger/hand, knee, wrist, elbow, foot/ankle, eye, ear, head/face, torso, pelvis/hip, or other). For each selected injury site, respondents were asked to identify the cause (check all that apply: missing or defective PPE, missing or inadequate safety infrastructure, work area not clear or closed off, too heavy loads, overworked, or other), as well as if their frequency of PPE use had changed as a result of the injury(s).

2.3.4 Data Reduction and Analysis

Only survey questions related to study purposes were analyzed. Additionally, all purposes were regarded as exploratory. Obvious outlying write-in responses were excluded, when applicable, such as when a participant misinterpreted a question. For example, when asked to report the average number of boulders set per day, 99.3% of respondents reported 15 boulders per day or less while only 1 respondent reported 35 boulders per day. This response is a clear outlier based on all existing routesetting literature, industry standards, and other data, and the respondent very clearly misinterpreted the prompt. This respondent's write-in response was not included in analysis or reporting. Data from the same respondent that properly answered other questions were preserved, such as age, ethnicity, years routesetting, etc.

Several respondents also wrote-in responses to questions about wall height using the wrong unit of measurement (feet instead of meters). These responses were converted to the correct unit of measurement to properly answer the question and this data was analyzed.

2.4 Results

2.4.1 Survey Response

178 people consented and participated in the survey. Of those, 133 (75%) respondents finished the survey. 56% of responses occurred within the first 48 hours of initial posting. Additional responses occurred in conjunction with the time

of reposts and email distributions with very few responses occurring between distributions (Figure 1). The average response time was 11:36 minutes.

As the survey was distributed using broad distribution methods (i.e., snowball sampling), a true response rate cannot reliably be calculated. For the pilot study conducted by the author,¹⁰ a similar recruitment strategy was used and elicited 117 responses during 1 week of being open. The Route Setters International Survey from 2018,⁹ distributed using the corporate resources of Vertical Life, yielded 976 responses during an unknown time period.

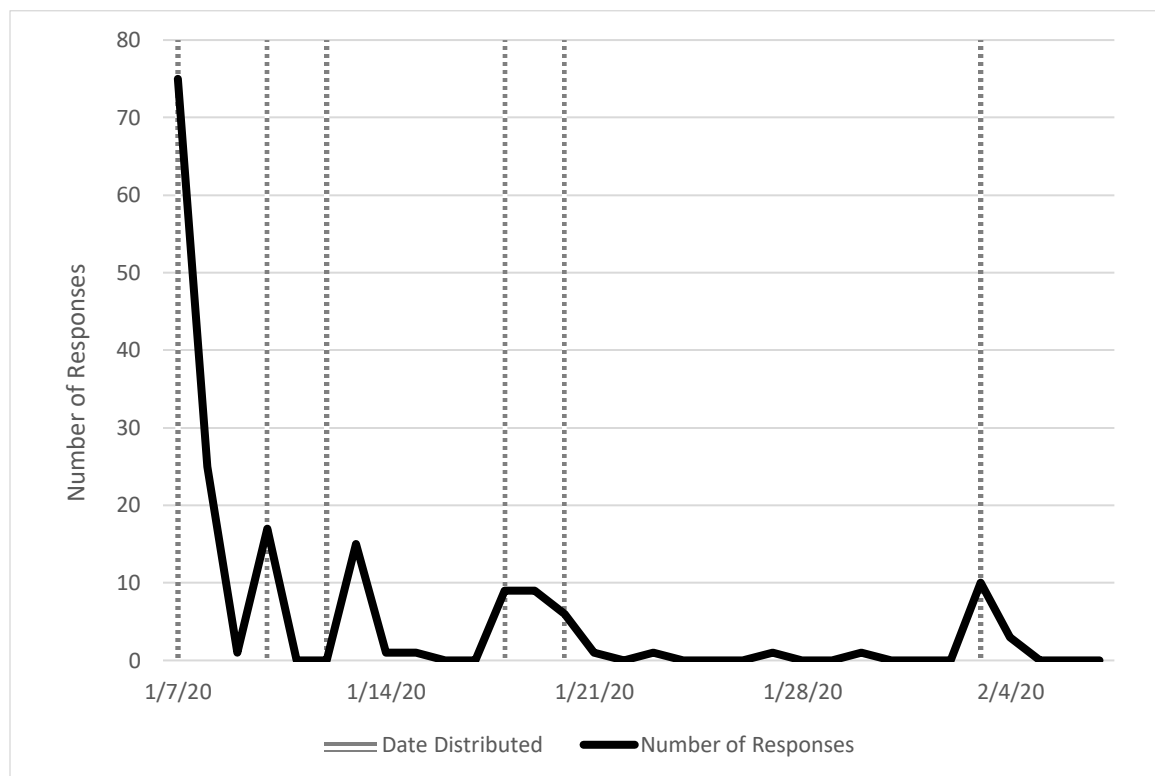


Figure 1: Distribution Schedule and Response timeline

2.4.2 Survey Questions

Demographic Questions

The largest proportion of respondents (46%; $n=78/168$) indicated they were 25-34 years old. Another 32% ($n=53/168$) of respondents reported being 18-24 years old (Figure 2). Over 85% ($n=144/168$) of respondents reported their ethnicity as “White” followed by “Asian” at 8% ($n=14/168$) (Figure 3). The majority of respondents were male (78%; $n=131/168$) (Figure 4). Respondents reported an average of 1.8 days off from physical activity on average each week (Figure 5) and an average of 11 years rock climbing (Figure 6).

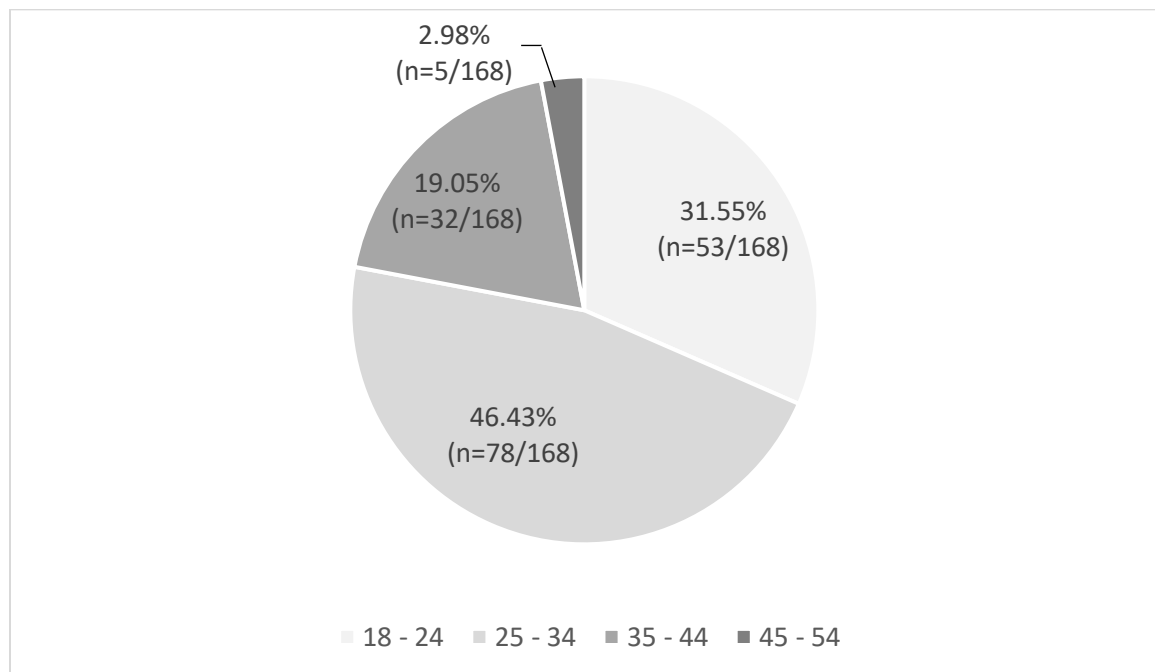


Figure 2: Distribution of Age of Respondents

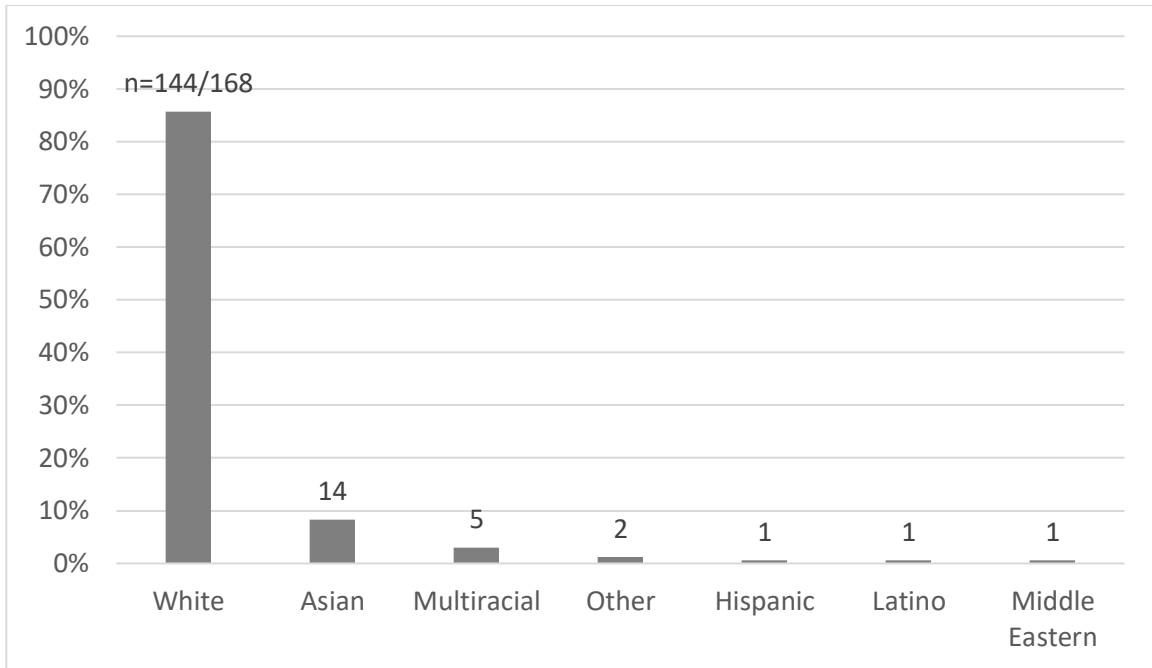


Figure 3: Distribution of Ethnicity of Respondents

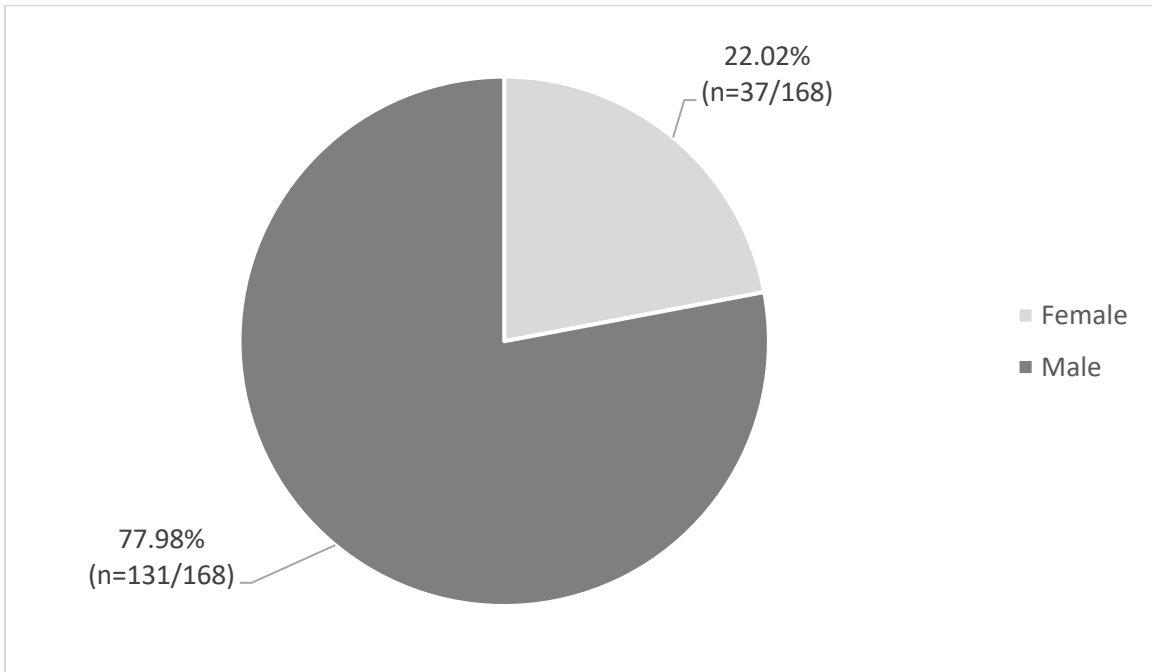


Figure 4: Distribution of Gender of Respondents

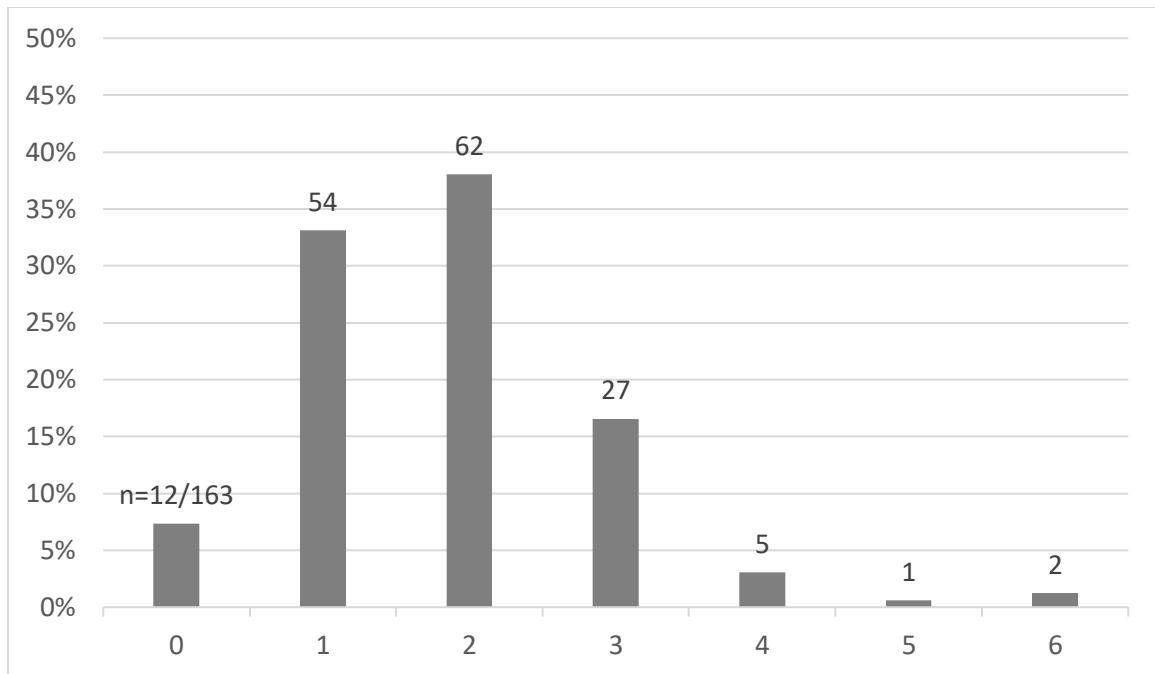


Figure 5: Complete days off from physical activity per week

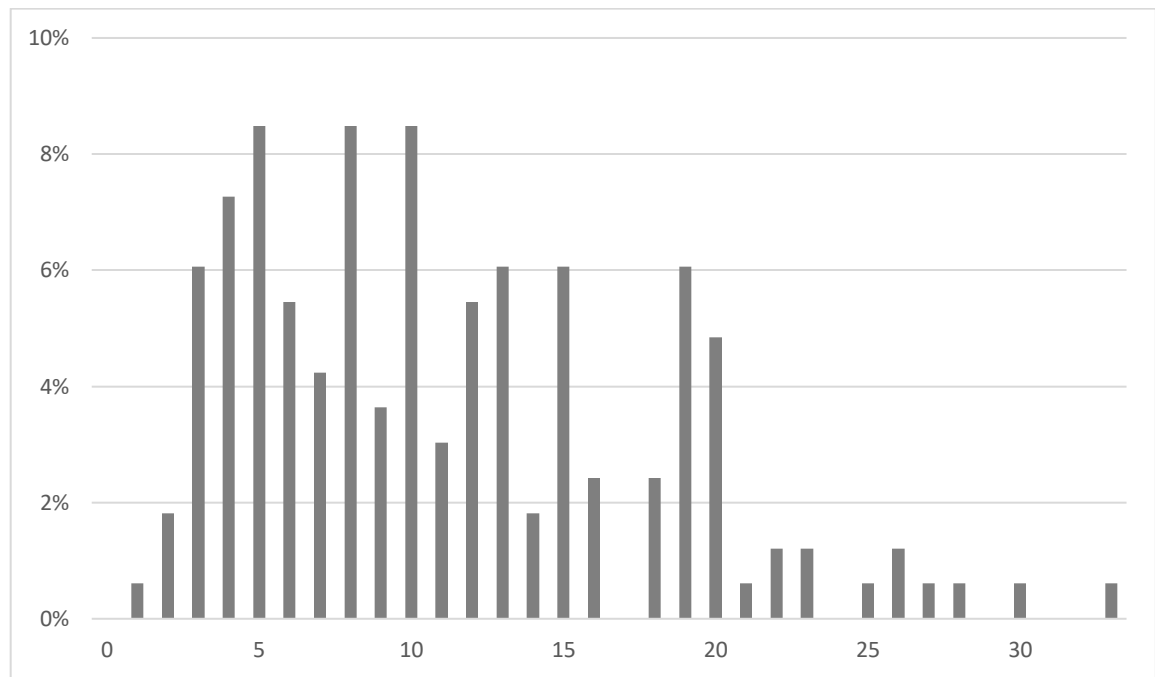


Figure 6: Distribution of years rock climbing (n=165)

Scope of Labor

Respondents reported they had been routesetting for an average of 5.5 ± 4.8 years (range: 1-25 years) (Figure 7). Most respondents (63%; $n=100/160$) reported working for a single rock climbing facility, while 23% ($n=37/100$) reported working for 2 facilities (Figure 8). Over $\frac{3}{4}$ (78%; $n=141/197$) of all reported climbing facilities were commercial gyms (Figures 9). Most respondents (70%) set between 1 and 4 days each week; an equal number of respondents set 1-2 days per week, 2-3 days per week, and 3-4 days per week (23% each, $n=35/150$) (Figure 10). More than $\frac{1}{3}$ (38%; $n=56/149$) of respondents set an average of 40-60% routes and 40-60% boulders (Figure 11b). Almost one quarter of respondents (19%; $n=29/149$) reported setting just boulders (Figure 11a). Most participants (77%, $n=113/147$) set mostly in a commercial setting with some competition setting as well (Figure 12). Just under half of respondents (45%; $n=66/147$) set commercially 90% of the time and for competition 10% of the time.

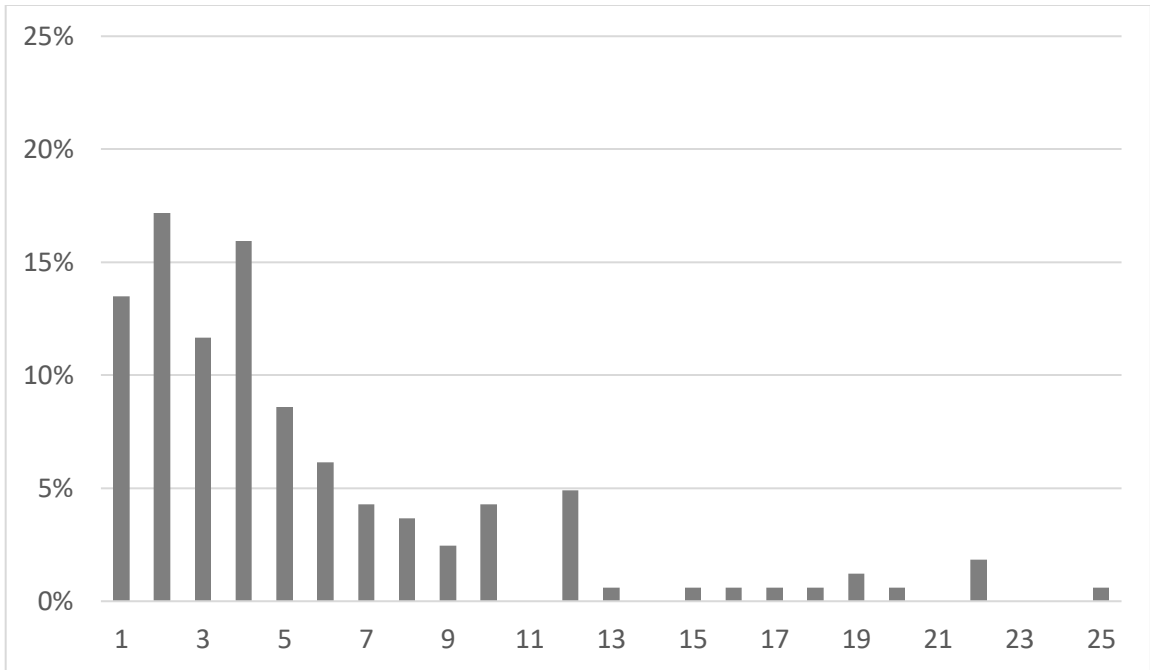


Figure 7: Distribution of years routesetting (n=163)

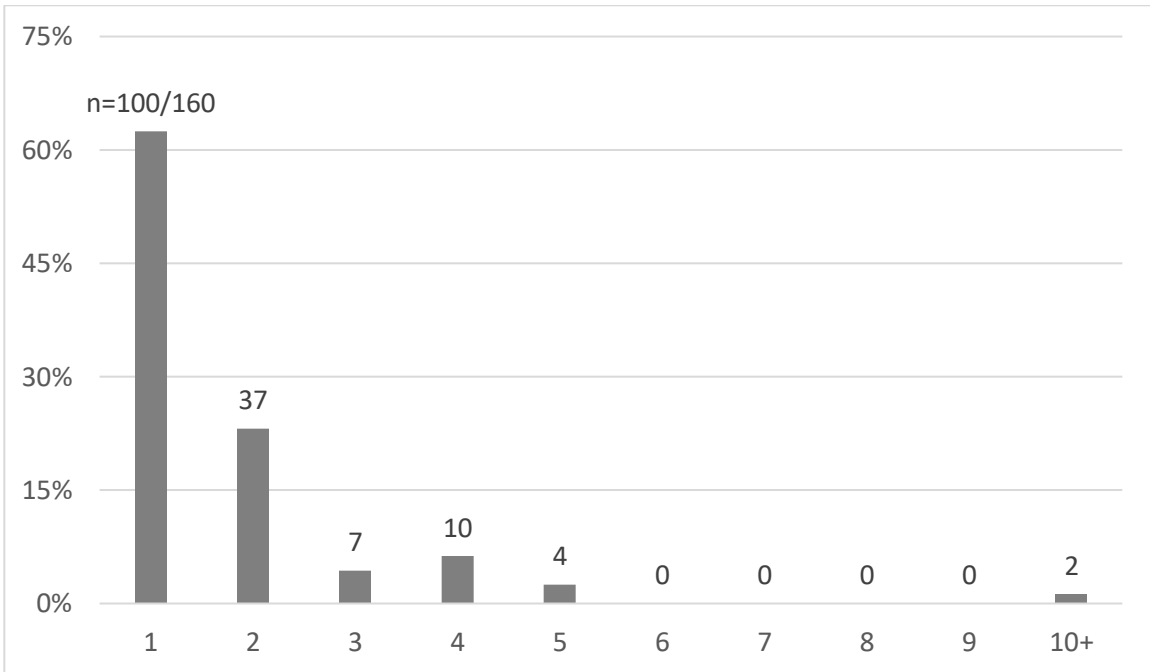


Figure 8: How many climbing facilities do you currently set for?

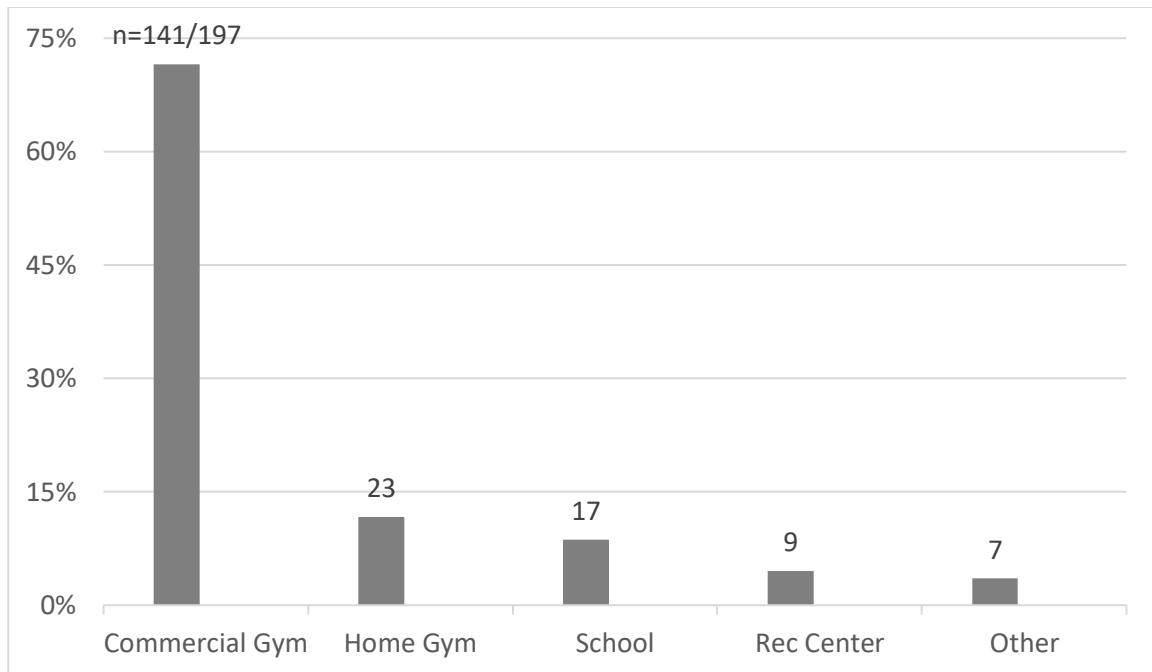


Figure 9: Where do you currently routeset?

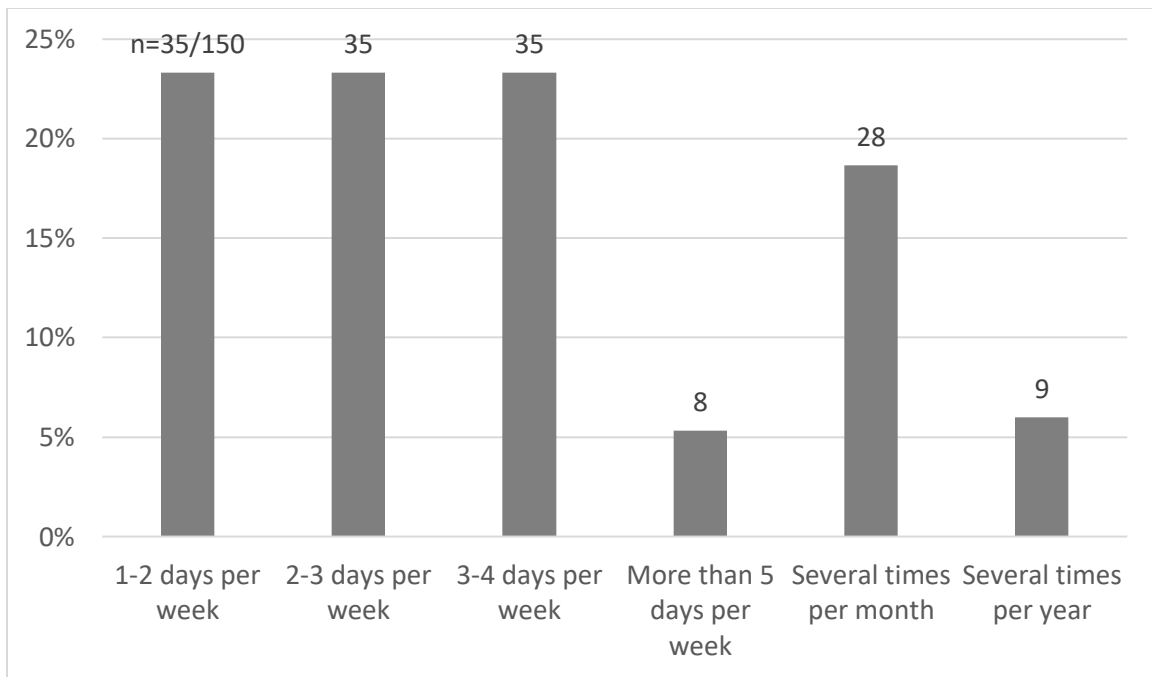


Figure 10: How often do you set?

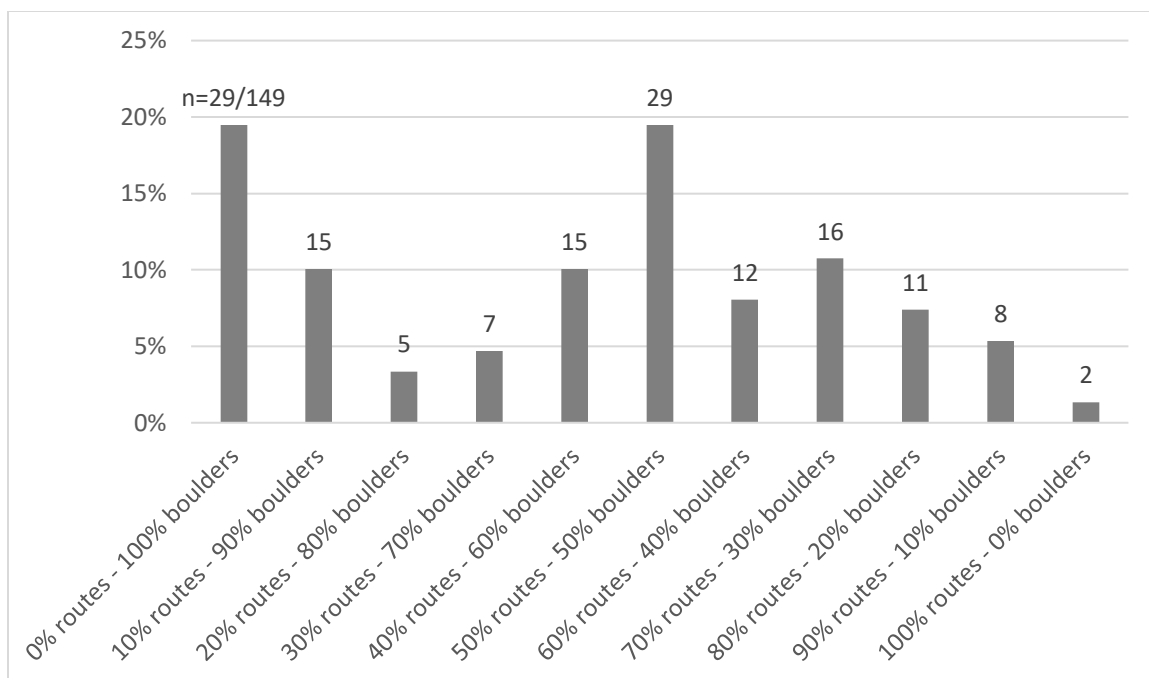


Figure 11a: How often do you set routes versus boulders?

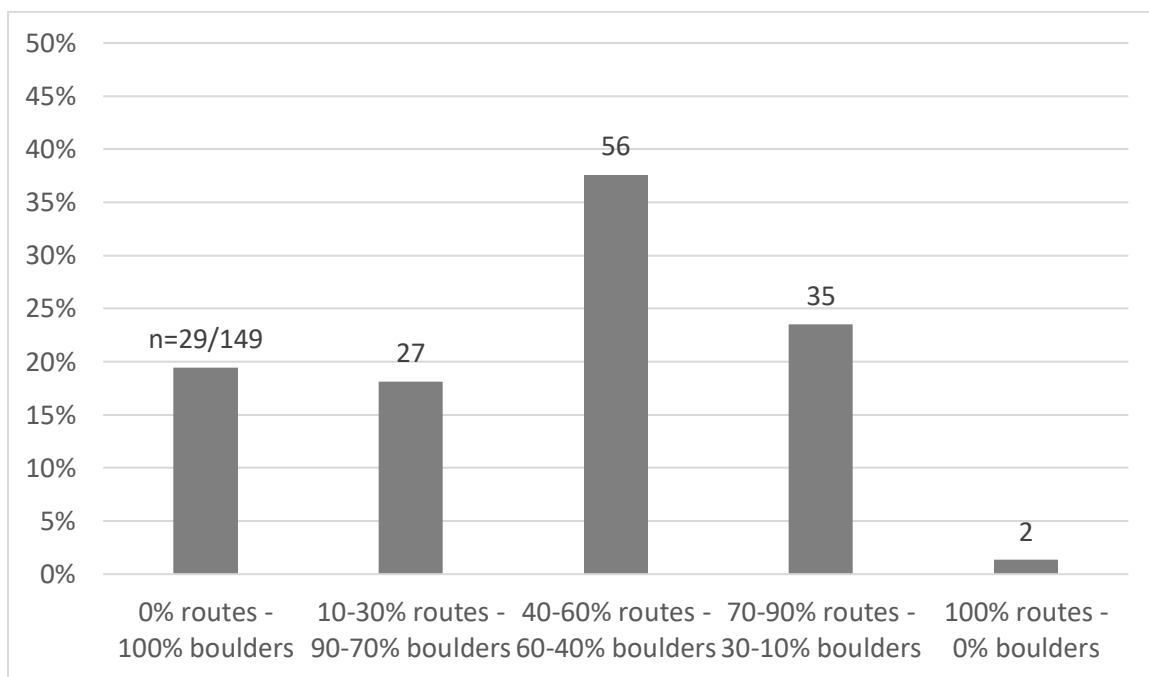


Figure 11b: How often do you set routes versus boulders? (grouped)

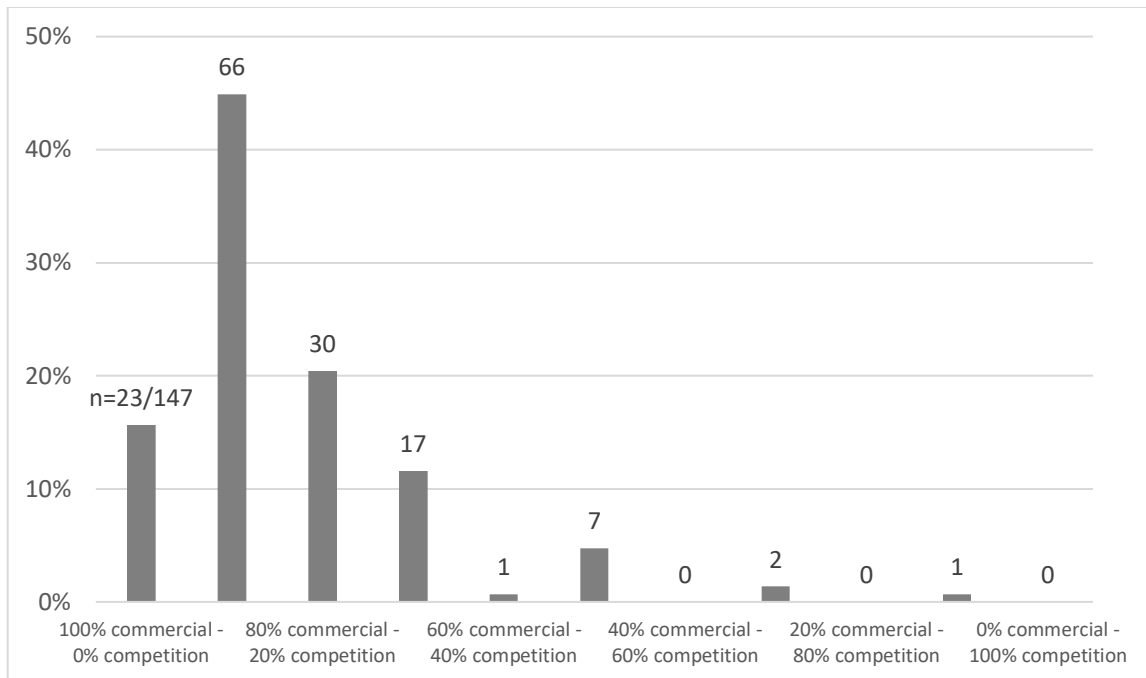


Figure 12: Do you mostly set commercially or for competition?

The average number of boulders respondents reported they set per day is 5.7 ± 3.7 , on a wall of an average height of 4.3 ± 1.0 meters (Figures 14 and 15). The average number of routes respondents reported they set per day is 2.6 ± 1.6 , on an average wall height of 13.2 ± 4.2 meters (Figures 16 and 17). When asked what activities apply to regular work, all respondents (100%; $n=143/143$) reported routesetting, 99% ($n=141/143$) reported testing/forerunning, 90% ($n=129/143$) reported hold stripping, 71% ($n=102/143$) reported hold sorting, and 66% ($n=94/143$) reported hold cleaning (Figure 18). 16 respondents (11%) also reported administrative work under “Other” as part of their regular routesetting labor; these tasks include training new setters, scheduling, and management.

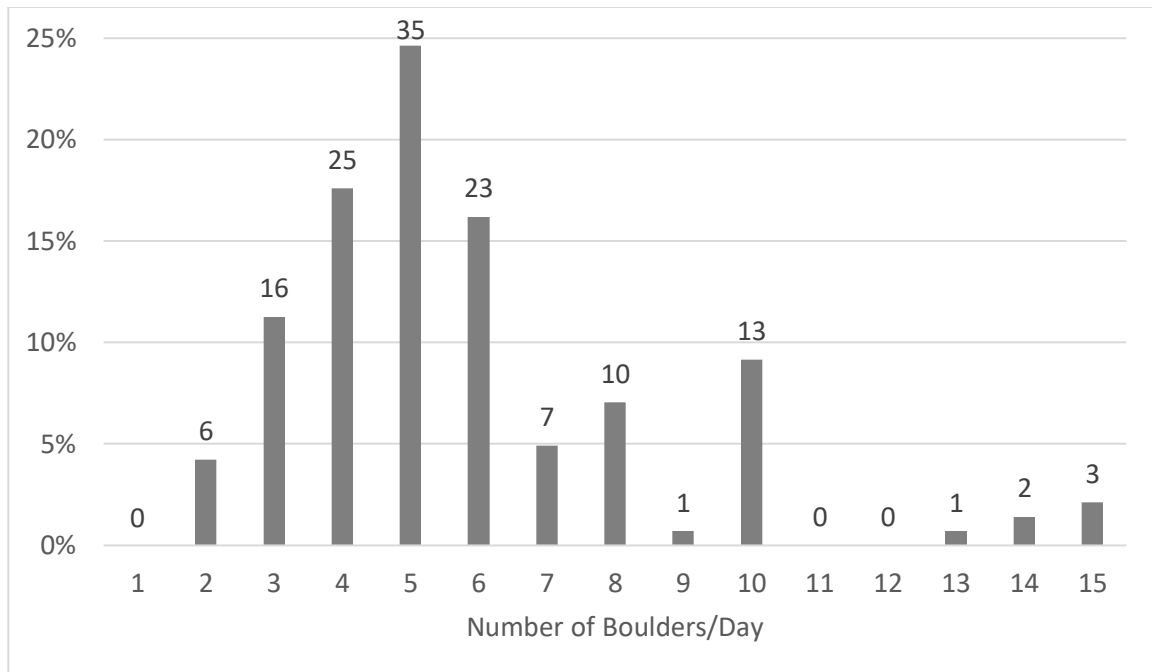


Figure 13a: On, average, how many boulders do you set in a day?

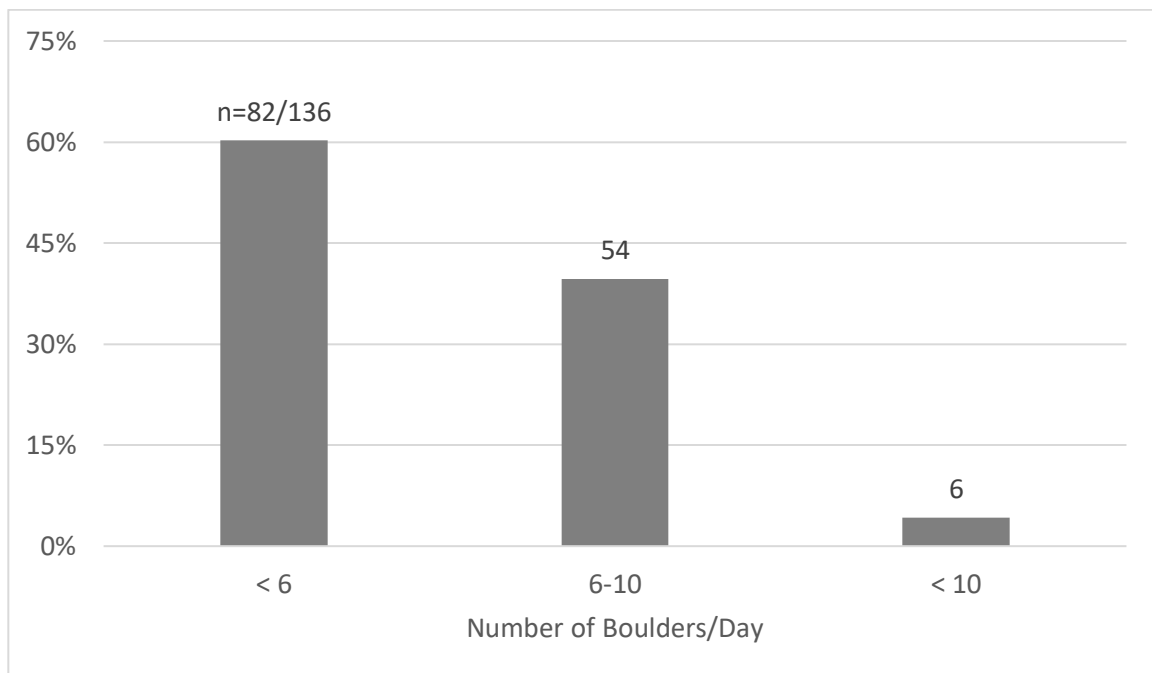


Figure 13b: On, average, how many boulders do you set in a day? (grouped)

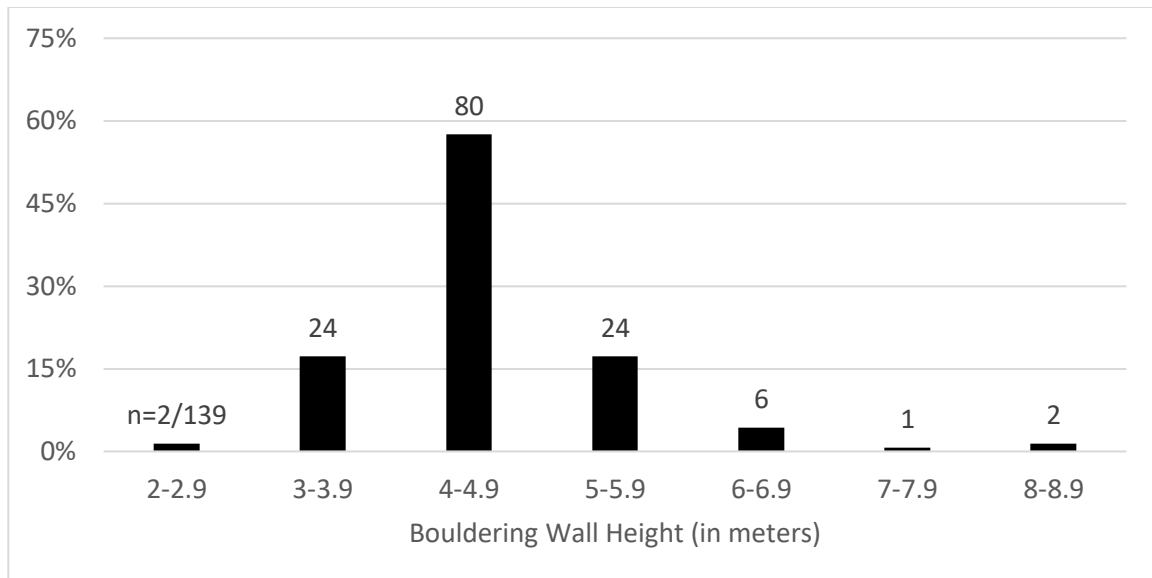


Figure 14: On average, how tall are your bouldering walls?

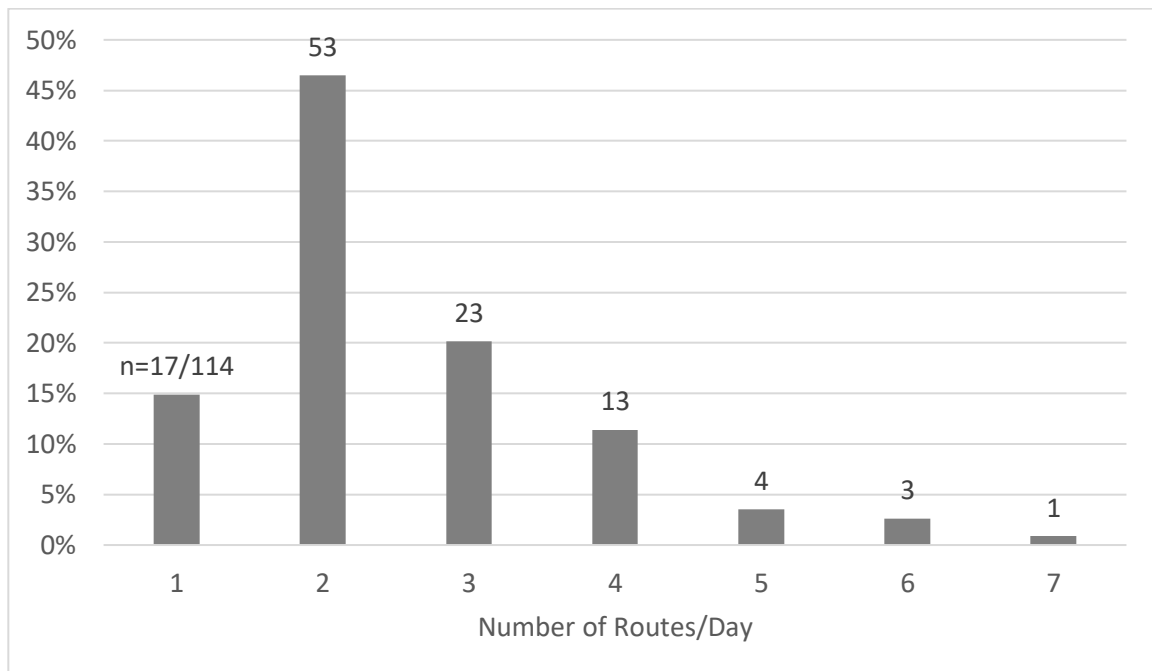


Figure 15: On average, how many routes do you set per day?

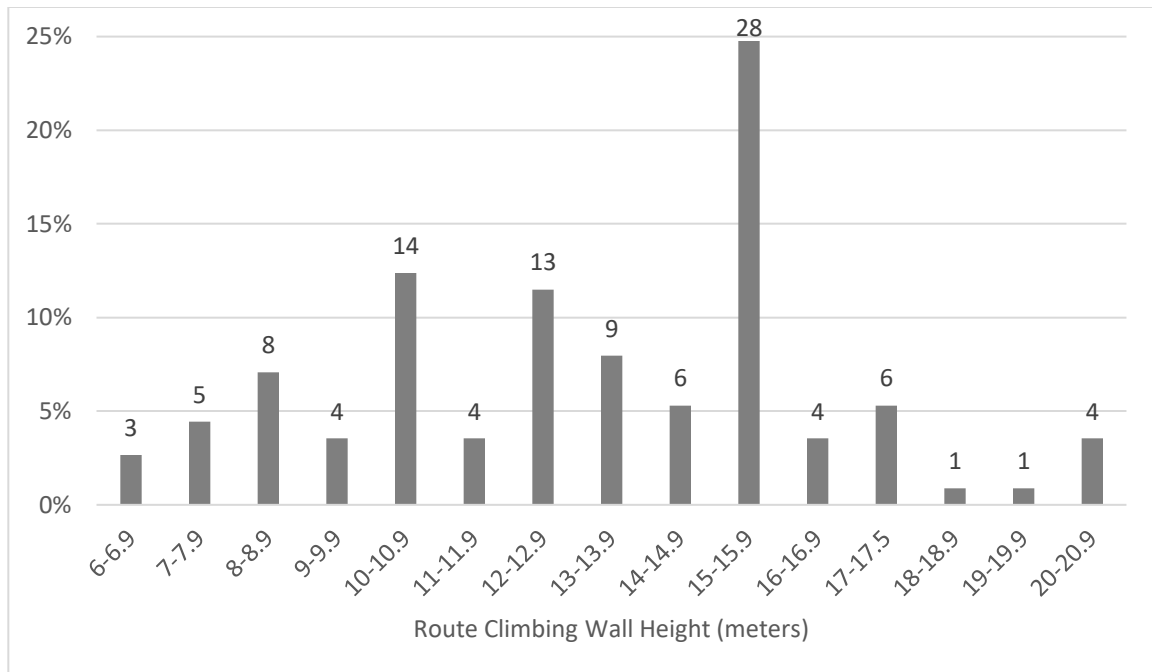


Figure 16: What is the average height of your route climbing walls? (n=110)

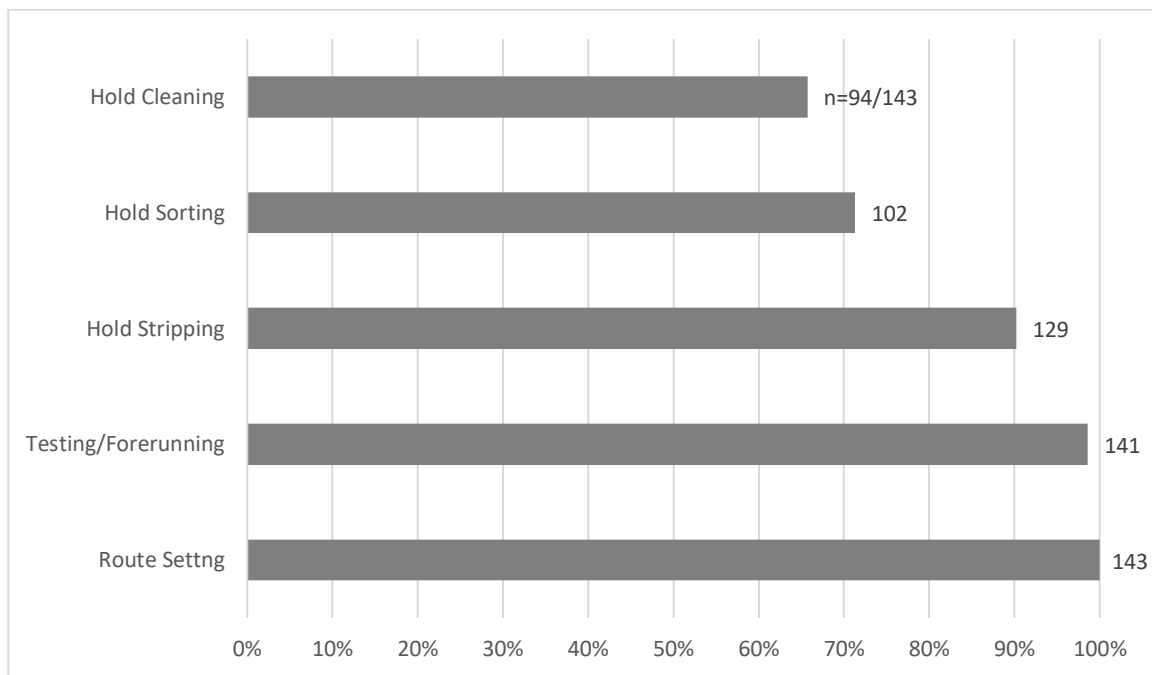


Figure 17: What activities apply to your work? (check all that apply)

Respondents were also asked to report specific labor tasks associated with regular routesetting by checking all that apply from a provided list. Almost all respondents (96-100%; 137-143/143) reported climbing, operating power tools, lifting heavy equipment, and ascending/descending ladders as part of their regular labor. Over $\frac{2}{3}$ of respondents (71-73%; 101-104/143) also reported ascending a fixed rope, belaying, and hauling heavy equipment on a rope. Less than $\frac{1}{4}$ of respondents (22%; 32/143) reported operating a lift. Of the participants who selected tool use, 89% (n=127/142) reported using an impact tool to install bolts (Figure 20). Of these impact tool users, 62% (n=78/125) use a conventional impact while 33% (n=41/125) use an oil impulse/hydraulic impact tool (Figure 21).

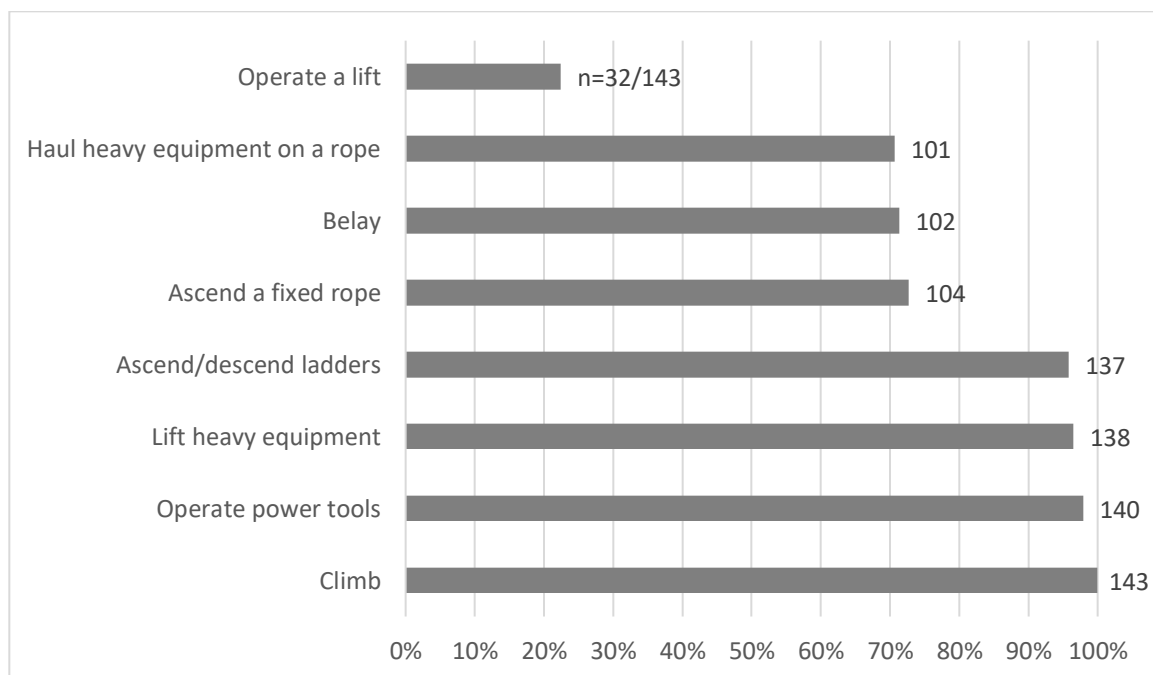


Figure 18: Do you do any of the following? (check all that apply)

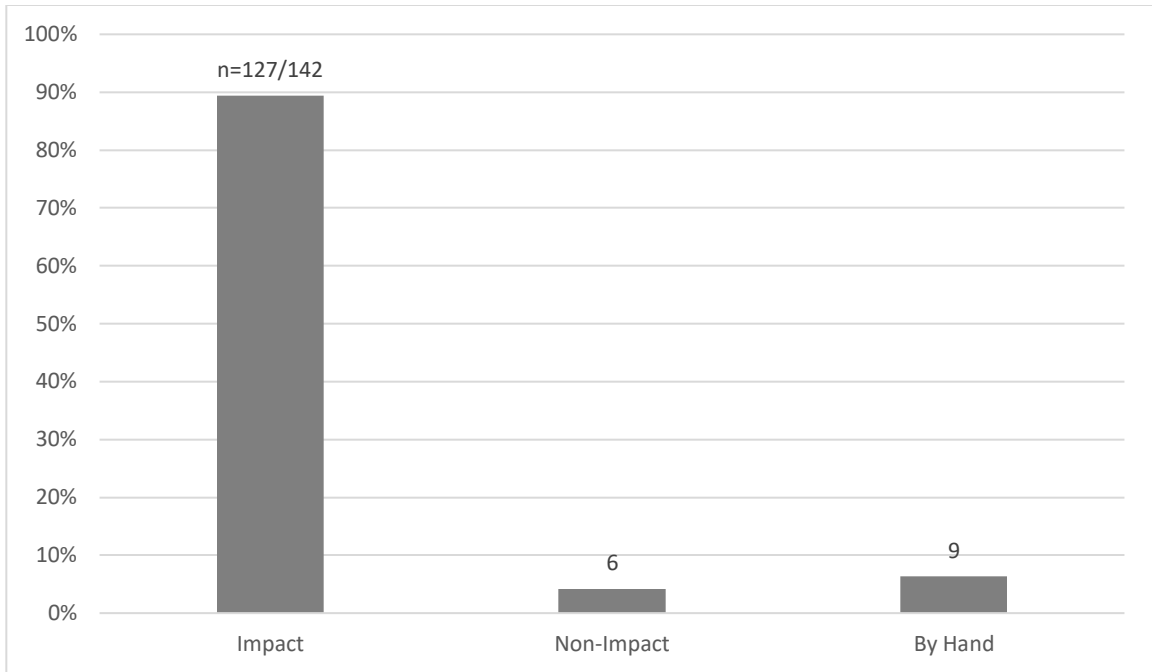


Figure 19: What tool do you use for bolts?

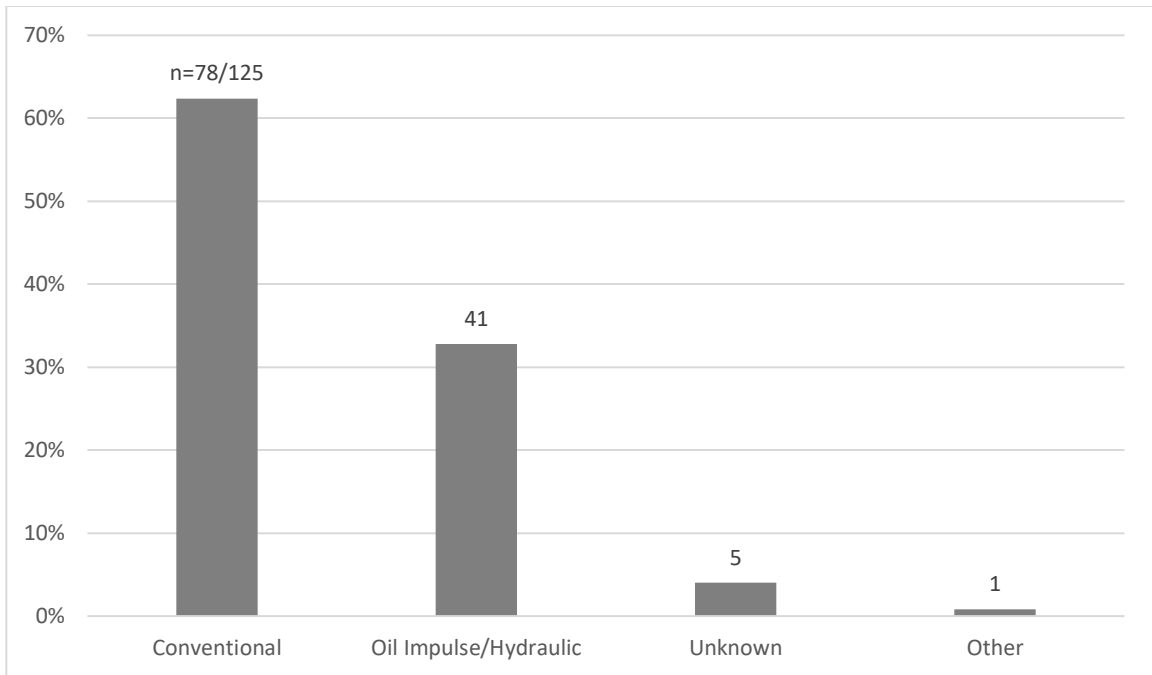


Figure 20: What type of impact tool do you use?

2.4.2 Injury History Questions

PPE use was measured in three categories: eye protection, ear protection, and head protection (helmet). 35% (n=51/141) of respondents reported always using eye protection while 21% (n=29/141) reported never doing so (Figure 22). Only 20% (n=26/130) of respondents reported always using ear protection while 39% (n=51/130) reported never doing so. Respondents who set routes were asked to report frequency of helmet use when doing so. Just 16% of respondents (n=18/110) always wear a helmet when setting routes, while 22% (n=24/110) rarely wear a helmet and 53% (n=58/110) never do so. 36% (n=50/140) of participants reported warming-up before performing routesetting labor while 64% (n=90/140) did not (Figure 23).

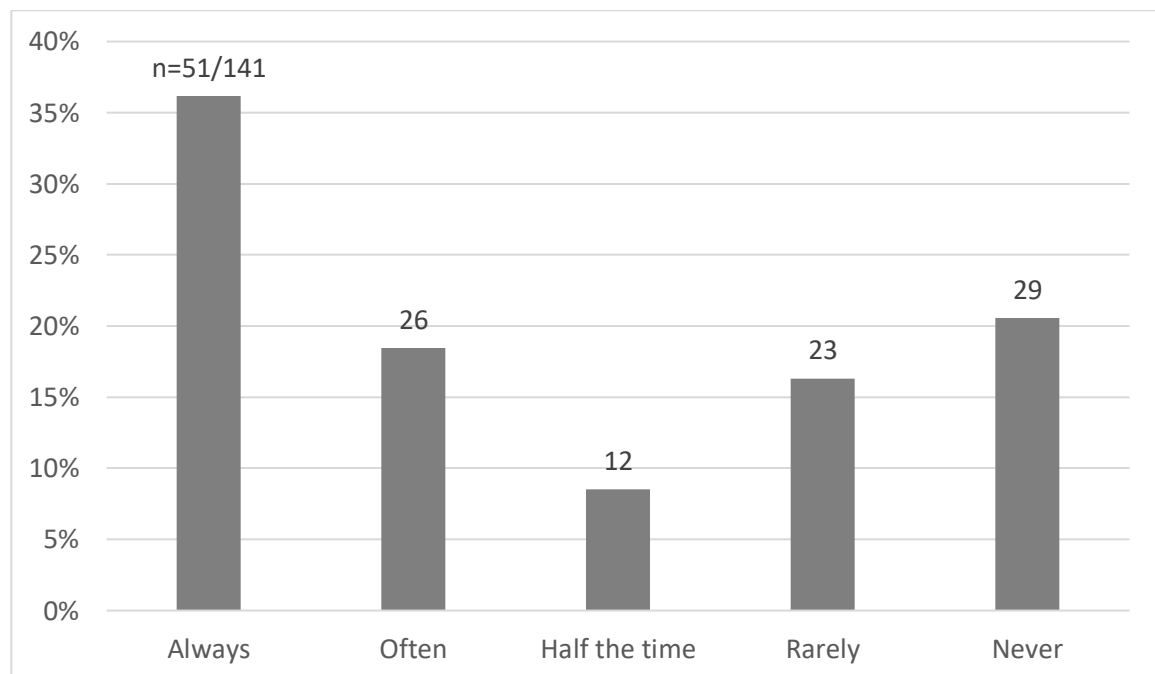


Figure 21: How often do you wear protective eye-wear?

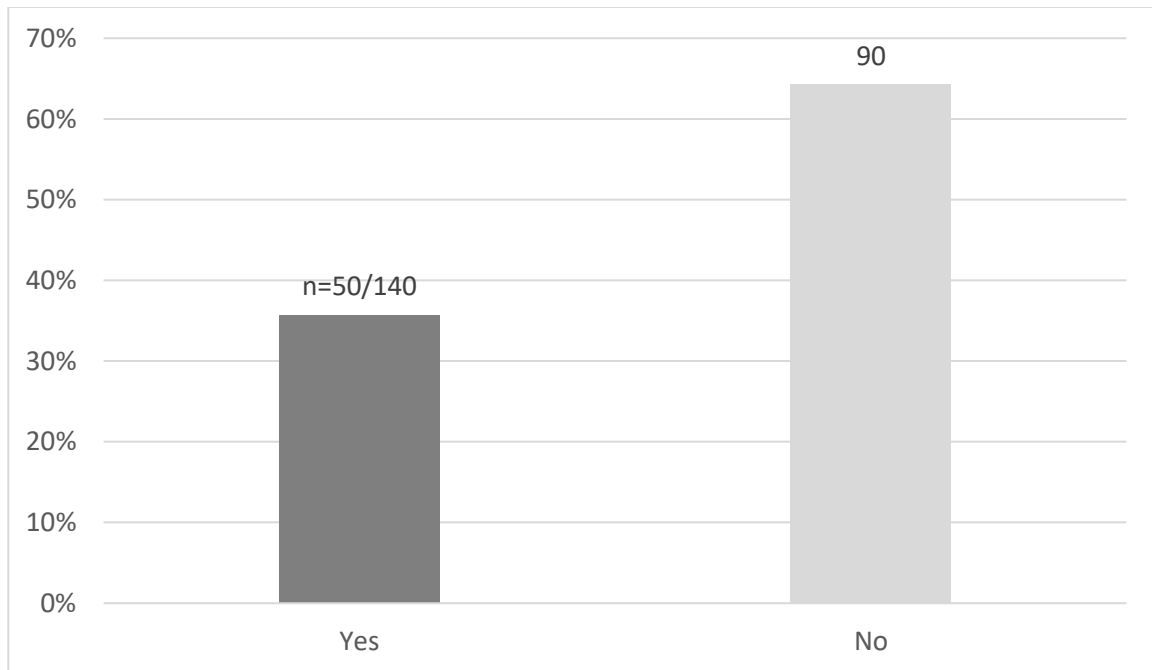


Figure 22: Do you warm up before performing routesetting labor?

Almost two-thirds of respondents (64%; $n=87/135$) reported experiencing an injury related to routesetting at some point during their routesetting career (Figure 24). Of those who reported an injury, the average number of career injuries was 4.2 ± 3.3 . Almost half (45%; $n=60/134$) reported experiencing at least one injury in the past year and, of those who reported an injury in the past year, the average number of injuries was 2.0 ± 1.5 . The most reported injury site was to the finger/hand; 21% ($n=60/288$) of respondents reported an injury at this site. Respondents also frequently reported shoulders, back, wrist, elbow and eye as a site of injury, making up the top six injury sites. (Figure 25). The leading cause of injury for the top five injury sites was “Overworked”; among fingers/hands, shoulders, backs, wrists, and elbows, 50-60% of respondents reported overworked as the cause of injury (Figures 26a-e). The leading cause of injury for the eye, however, was missing or defective PPE, reported by 78% ($n=21/27$) of

respondents who reported eye injuries (Figure 26f). Of those who reported an eye injury, 81% (n=21/26) were not wearing eye protection when they sustained the eye injury(s) and 85% (n=22/26) reported being more likely to use PPE after their injury(s) (Figures 27 and 28).

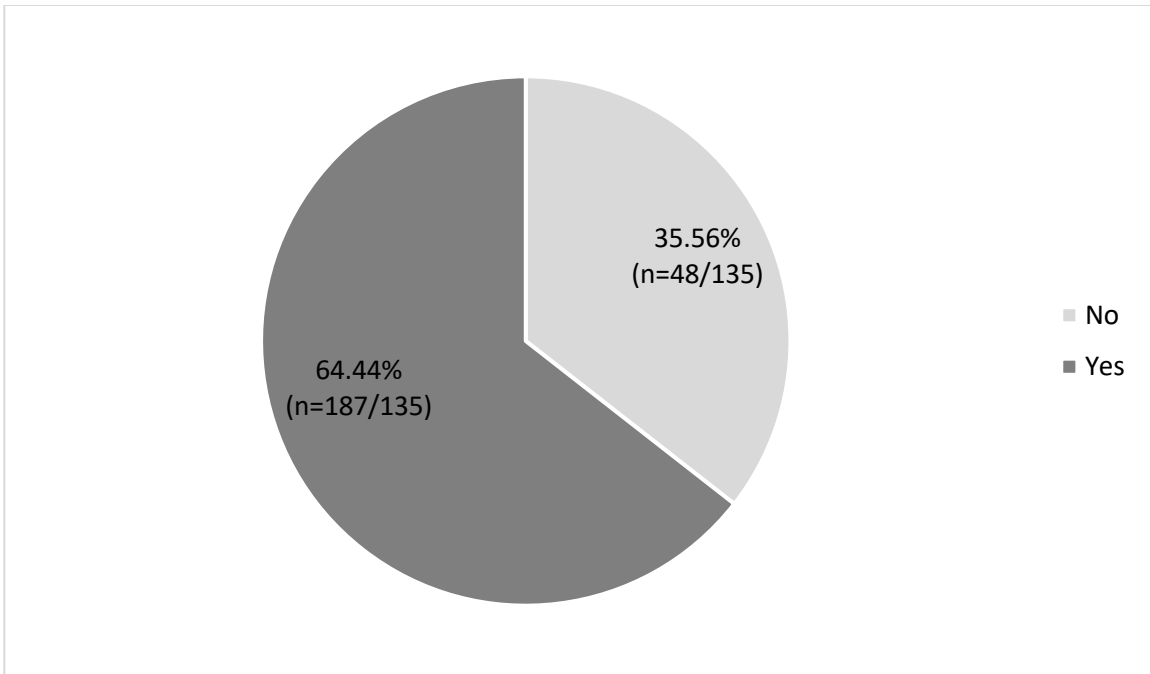


Figure 23: Have you ever experienced an injury related to routesetting?

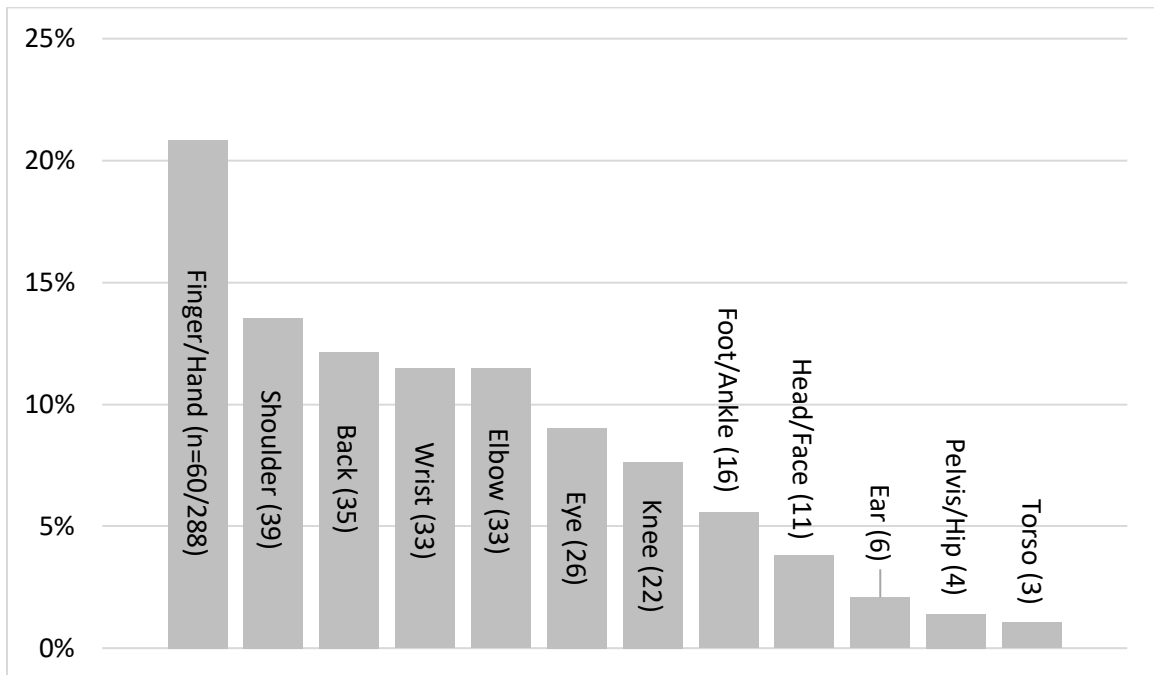


Figure 24: Have you experienced an injury to any of the following? (n=288)

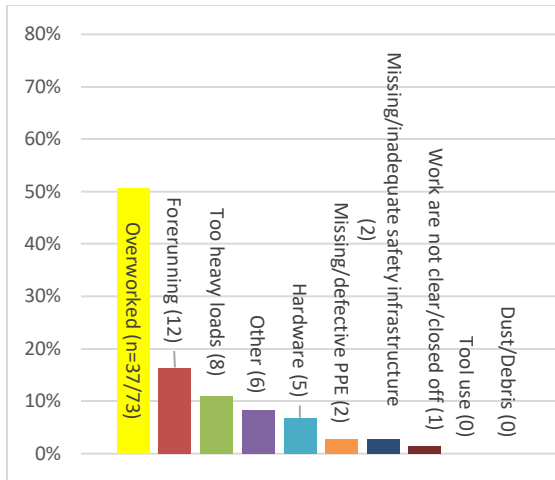


Figure 25a: Cause of Finger/Hand Injury

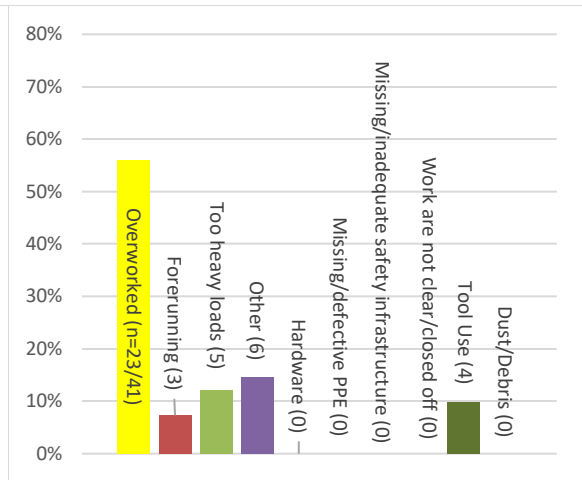


Figure 25d: Cause of Wrist Injury

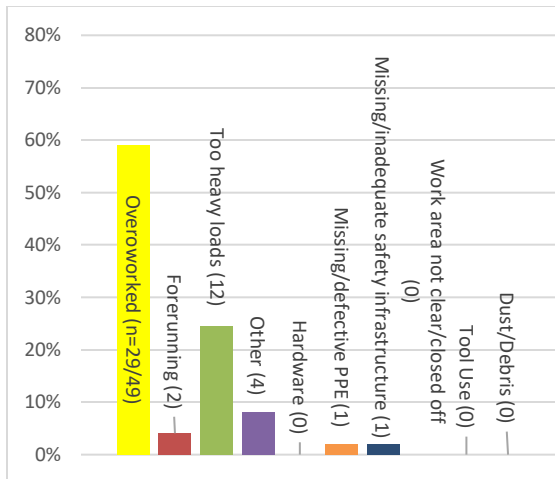


Figure 25c: Cause of Back Injury

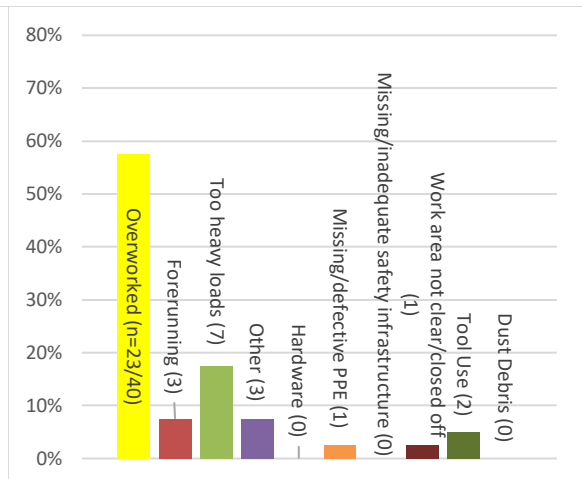


Figure 25e: Cause of Elbow Injury

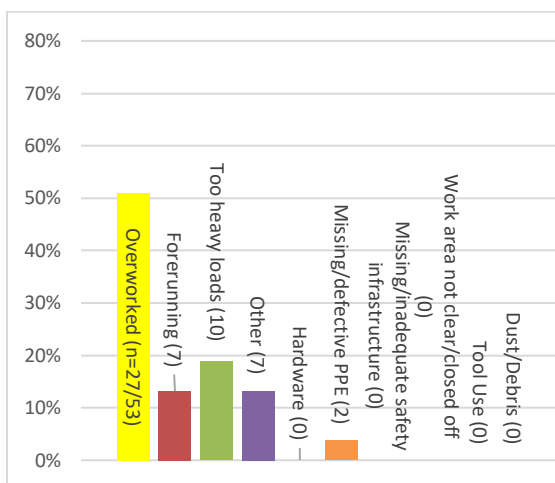


Figure 25b: Cause of Shoulder Injury

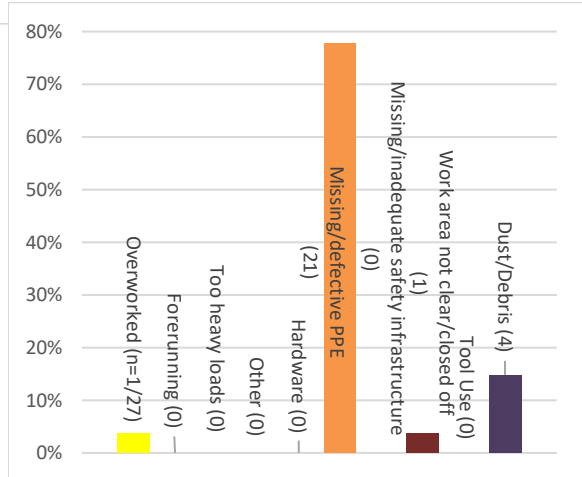


Figure 25f: Cause of Eye Injury

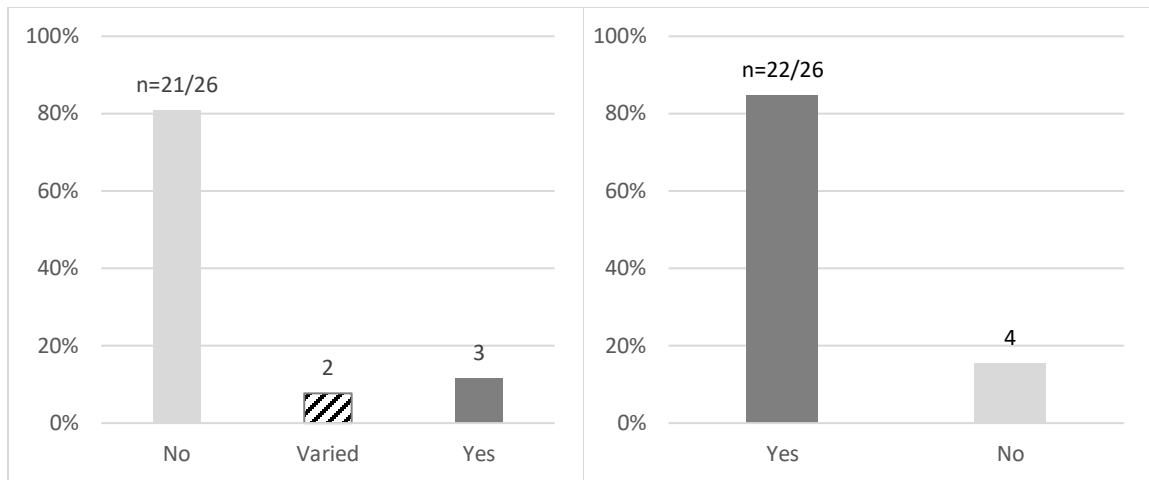


Figure 26: Were you wearing eye protection when you sustained an eye injury?

Figure 27: Are you more likely to use PPE after your eye injury?

2.5 Major Findings

2.5.1 Why are routesetters getting injured?

The primary purpose of this study was to evaluate the prevalence of injury and explore potential cause(s) of injury in routesetters. For each reported injury site, respondents were asked to identify the cause of their injury(s) from the following options: overworked, too heavy loads, missing/defective PPE, missing or inadequate safety infrastructure, work area not cleared or closed off, or other. The top five injury sites accounted for 68% of all reported sites in the following order: finger/hand (21%), shoulder (14%), back (12%), wrist (11%) and elbow (11%). The leading cause of injury for all five groups was “overworked”: accounting for 50-60% of injuries among fingers/hands, shoulders, backs, wrists, and elbows (Figures 25a-f).

The second leading cause of injury varied. For finger/hand injuries, the second leading cause of injury was “forerunning” (16%) and the third leading cause was “too heavy loads” (11%). Among shoulder, back, wrist, and elbow injuries, the second leading cause was “too heavy loads” accounting for 12-24%. The third leading cause for these injuries was varied.

2.5.2 Does PPE use affect the likelihood of injury?

To determine if PPE use affects the likelihood of injury, we examined the relationship between eye injury and eye protection. We chose to report solely on PPE use in eyes based on our analysis for type of injury, in which eye injuries represented a significant portion of reported injuries (9%; 26/288) while ear (2%; 6/288) and head/face (4%; 11/288) injuries did not. Eyes were the sixth highest ranking injury site reported by participants, accounting for 9% of all reported injuries. The leading cause was missing or defective PPE, accounting for 78% of all reported eye injuries. Dust/debris in the eye accounted for 15% of these injuries. 85% of those who reported an eye injury reported being more likely to use PPE after the injury(s).

2.5.3 Does the scope of labor affect the likelihood of injury?

Scope of labor was primarily measured using three metrics: number of days setting per week, time spent setting routes versus boulders, and time spent setting commercially versus for competition. Regarding days worked per week, 86.21% of those who reported working 3-4 days per week reported an injury. This group reported the highest occurrence of injury, barely ahead of those who

worked five or more days per week who reported an injury occurrence of 85.71% within the group (Figures 29a and b). The group that reported working 2-3 days per week reported the third highest occurrence of injury, with 75.76% of respondents in this group reporting an injury. Participants who reported working the lowest frequency (several times per year) reported the fourth highest occurrence of injury at 57% (Figure 34f). The groups who reported working 1-2 days per week and several times per month reported the lowest occurrence of injury at 48% and 42%, respectively (Figures 32d and 33e).

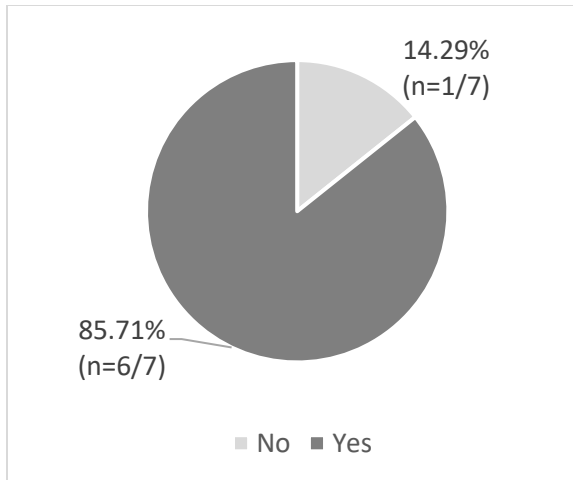


Figure 28a: Injury occurrence among those who work more than 5 days per week

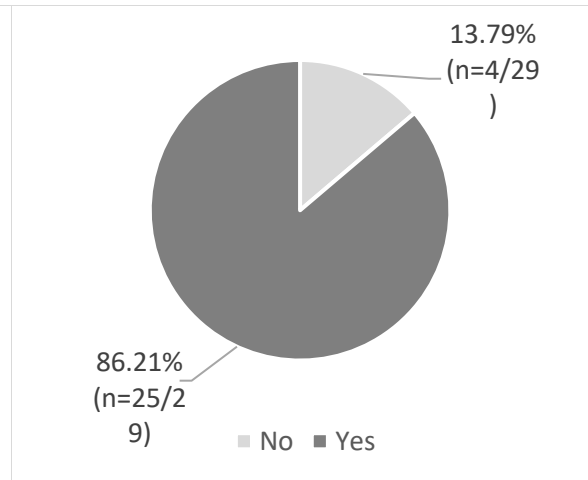


Figure 28b: Injury occurrence among those who work 3-4 days per week

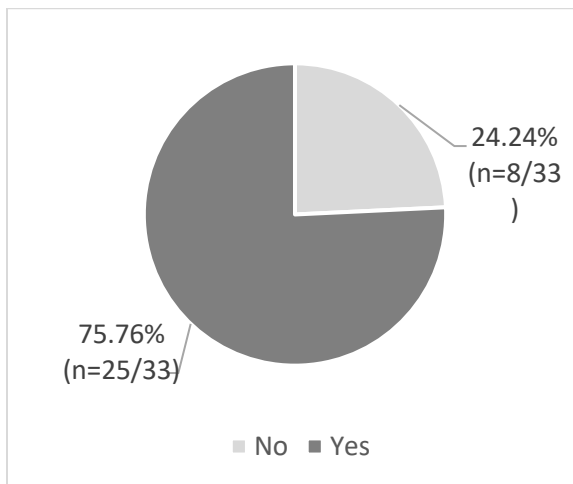


Figure 28c: Injury occurrence among those who work 2-3 days per week

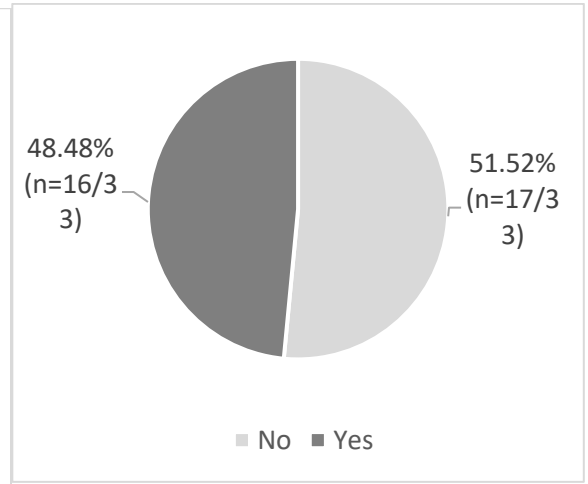


Figure 28d: Injury occurrence among those who work 1-2 days per week

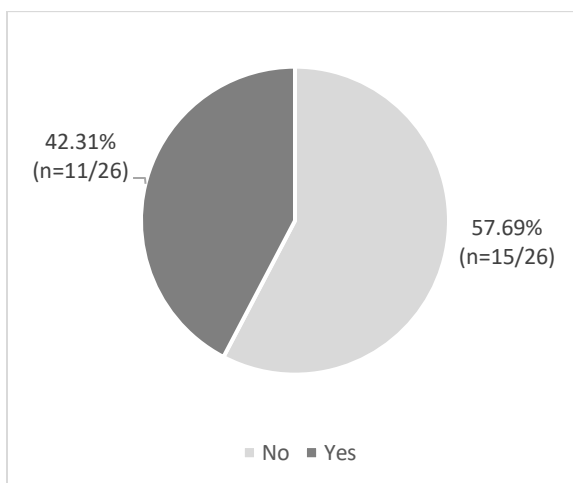


Figure 28e: Injury occurrence among those who work several times per month

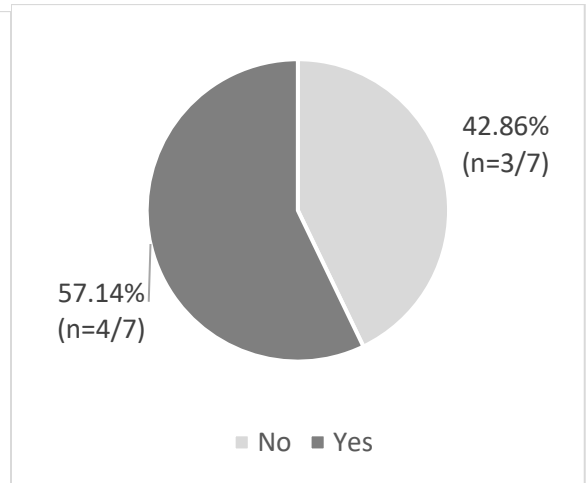


Figure 28f: Injury occurrence among those who work several times per year

Regarding time spent setting routes versus boulders, the group that reported setting all routes (100%) reported the highest occurrence of injury (100%; $n=2/2$) (Figure 30b). Of the group that reported setting some routes (10-30%) and mostly boulders (70-90%), 77% reported experiencing an injury ($n=20/26$). This group was very closely followed by those who reported setting about equal amounts of both routes (40-60%) and boulders (40-60%), of whom 76% ($n=38/50$) reported experiencing an injury related to routesetting (Figure 2.6.9). The lowest occurrence of injury was among the group setting all boulders (100%) and the group setting mostly routes (70-90%) and some boulders (30-10%) (48% [$n=12/25$] and 47% [$n=15/32$] respectively) (Figures 30a and 30d).

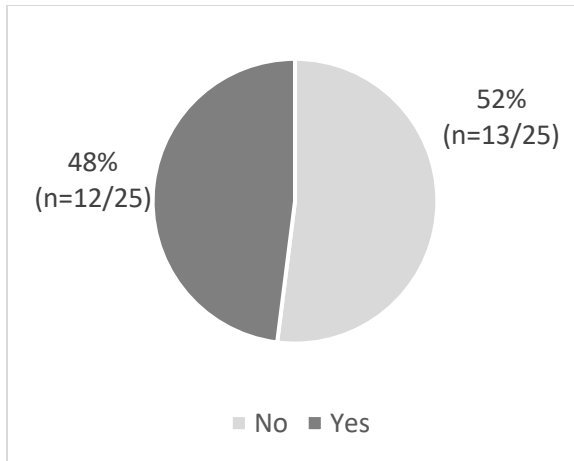


Figure 29a: Injury occurrence among those setting all boulders (0% routes and 100% boulders)

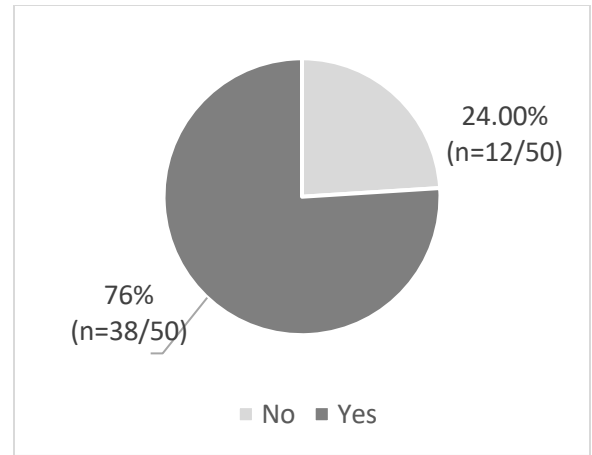


Figure 29c: Injury occurrence among those setting about equal amounts routes (40-60%) and boulders (40-60%)

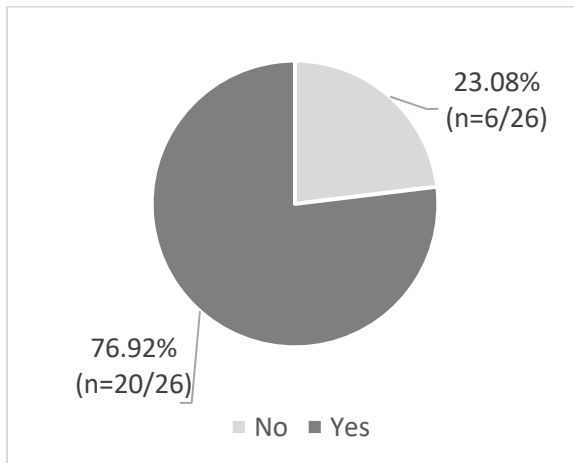


Figure 29b: Injury occurrence among those setting some routes (10-30%) and mostly boulders (90-70%)

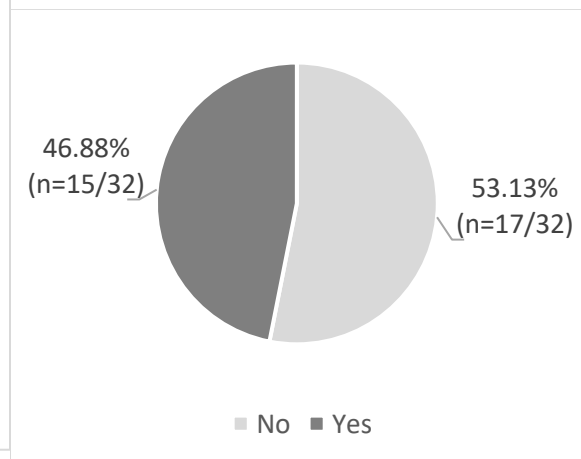


Figure 29d: Injury occurrence among those setting mostly routes (70-90%) and some (30-10%)

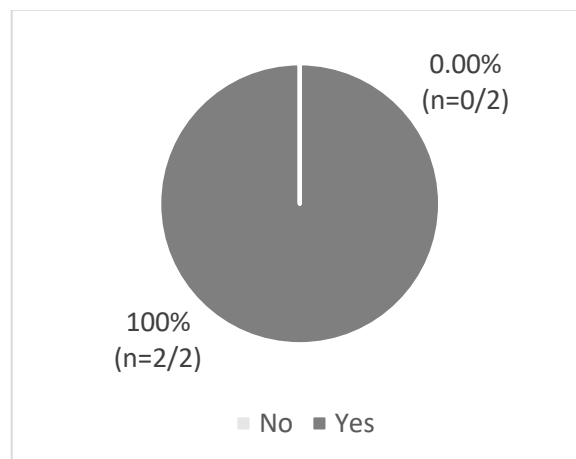


Figure 29e: Injury occurrence among those setting all routes (100%) routes and no boulders (0%)

Regarding time spent setting commercially versus for competition, the group setting 50% commercially and 50% for competition reported the lowest occurrence of injury at 17% ($n=1/6$) (Figure 31e). The second lowest occurrence of injury was in the group setting 100% commercially at 35% ($n=7/20$) (Figure 31a).

The highest occurrence of injury was in the group that reported setting 80% commercially and 20% for competition (81%; $n=22/27$) (Figure 31c). The second highest occurrence of injury was in the group setting 70% commercially and 30% for competition at 73% ($n=11/15$) (Figures 31b) The third highest occurrence of injury was in the group setting 90% commercially and 10% for competition at 68% ($n=43/63$) (Figure 31d).

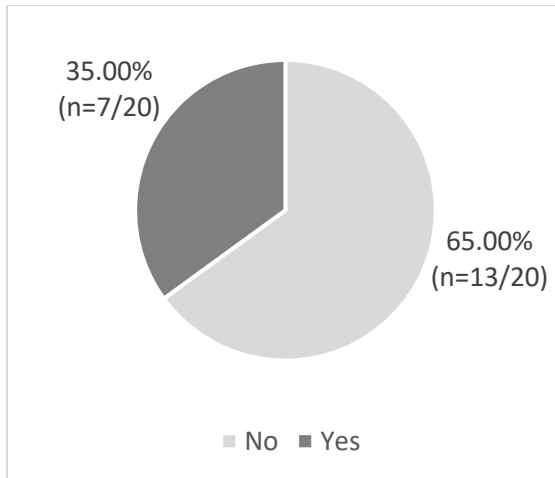


Figure 30a: Injuries among those setting 100% commercially and 0% for competition

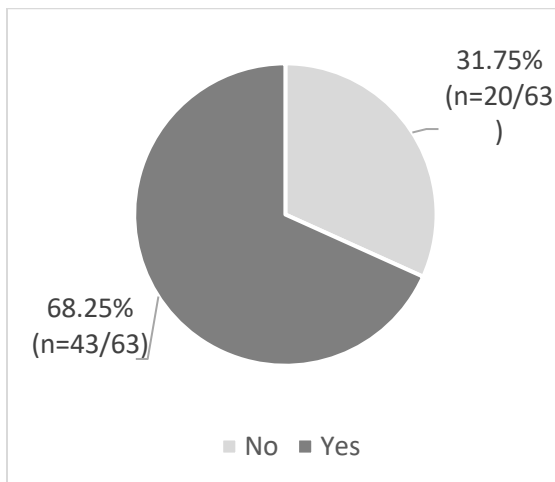


Figure 30b: Injuries among those setting 90% commercially and 10% for competition

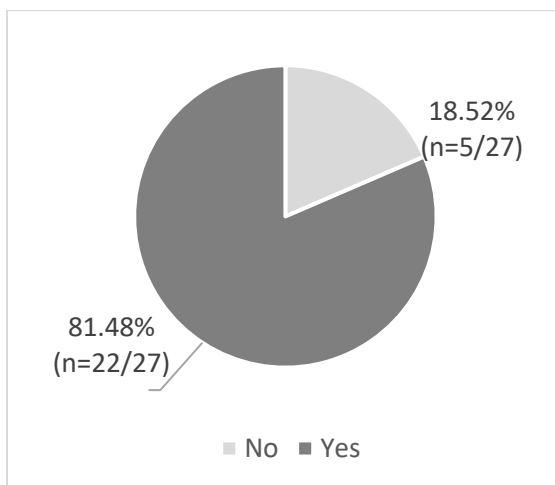


Figure 30c: Injuries among those setting 80% commercially and 20% for competition

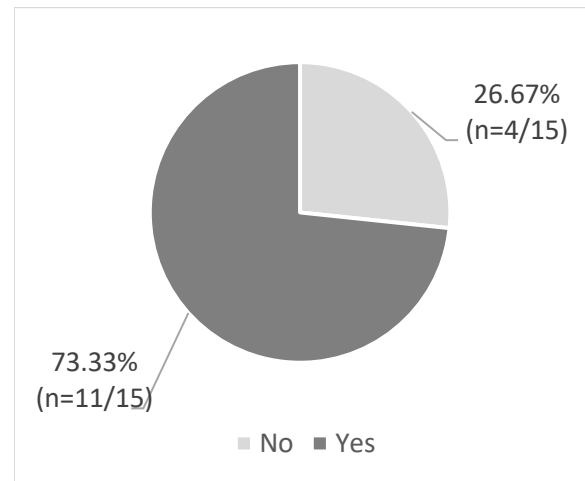


Figure 30d: Injuries among those setting 70% commercially and 30% for competition

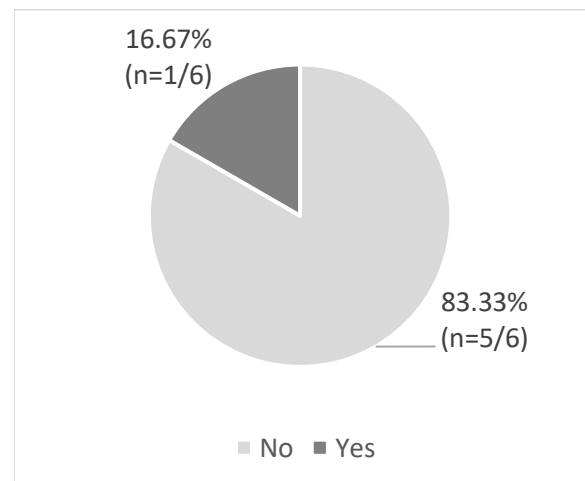


Figure 30e: Injuries among those setting 50% commercially and 50% for competition

2.6 Discussion

2.6.1 Demographics

The demographics of the survey population closely reflect those that have been reported in other studies, with only minor deviation. The majority of respondents (46%) were 25-34 years old, similar to the average age in the Dalby study of 29 years old¹⁰ and the average age in *Routesetters International* of 31 years old.⁹ In both this survey and the Dalby study, just over $\frac{3}{4}$ of respondents were male and just under $\frac{1}{4}$ of respondents were female. This represents a higher percentage of female routesetters compared to the 87% male and 13% female population reported by Zeidelhack et al. in *Routesetters International*.⁹ Furthermore, a 2019 survey by Gripped Magazine reported that 20-30% of routesetters in Canada were female.¹¹ Although there is slight variation between gender demographics of these different surveys, all of them indicate that the routesetting industry is predominantly male with less than $\frac{1}{4}$ of all routesetters reporting their gender as female.

Regarding length of climbing and routesetting career, the demographics were similar between these surveys. This survey and the Dalby study reported a slightly younger population of climbers and routesetters compared to the Vertical Life study. Respondents to both surveys¹⁰ had been climbing an average of 11 years compared to the 13 years reported in *Routesetters International*.⁹ Respondents to this survey had been routesetting for an average of 5.5 ± 4.8 years routesetting, slightly less than both the Dalby study (5.9 years) and the

Vertical Life study (7 years). Thus, this survey represents a slightly younger demographic of routesetters.

Although the Vertical Life study did not report the ethnicity of respondents, both this survey and the Dalby study¹⁰ reported nearly identical ethnicity demographics, with 85-87% of respondents reporting their ethnicity as “White,” followed by 6-8% reporting “Asian,” and 6-7% reporting “Other.” It should be noted that both of these surveys used nearly identical distribution avenues, and the similarities in the two demographics may be reflective of this distribution method. The Zeidelhack et al. study, on the other hand, used a wider range of distribution techniques but still yielded very similar demographic statistics.⁹

2.6.2 Are routesetters getting injured? Why are routesetters getting injured?

Almost two-thirds of respondents (64%) reported experiencing an injury related to routesetting at some point during their routesetting career.

Furthermore, almost half of respondents (45%) reported experiencing an injury within the past year. On average, respondents experience 3 injuries every 4 years, indicating that workplace injury is a recurring issue for routesetters.

Furthermore, in the Vertical Life survey *Route Setters International*,⁹ over half of respondents reported their perceived level of workplace safety was not high.

Thus, with both a high percentage of the population reporting regular workplace injury and a “less than high” level of workplace safety for most respondents, routesetting injuries are both common and, most likely, systemic.

Injury Sites

Routesetting injury sites closely represent the trends found in rock climbing, and construction injury trends are also present in this population. With 45% of routesetters reporting an injury within the past year, this closely matches the injury rate among rock climbers 50% of whom reported an injury within the previous 12 months in a study by Jones et al.³ According to that same study,³ and confirmed by findings from Grønhaug et al.,² the finger, shoulder, and elbow are the most frequently injured sites for climbers. All of these injury sites are within the top five highest most prevalent injuries among routesetters, representing a close overlap between injury trends among climbers and those found in routesetters (Figure 24). Therefore, routesetting injuries to the finger, shoulder, and elbow may be at least partially attributed to climbing.

Injuries to the back, fingers/hands, and wrists can be traced to construction injury trends. The back was the second highest site of injury among routesetters at 12%, and accounts for 8.5% of all construction injuries.¹ Furthermore, among construction workers, fingers, hands, and wrists account for 21% of all injured body parts.¹ Thus, injury sites among routesetters also reflect those trends found in construction work.

Finger/Hand Injuries

The leading site of injuries among routesetters was fingers/hand, and the leading cause of these injuries was being overworked, accounting for 51% of all reported finger/hand injuries. Forerunning accounted for 16% of finger/hand

injuries, representing another significant cause of injury. Finger injuries can be clearly associated with climbing, which places uniquely high loads on the finger flexor tendons¹² but is likely also a product of construction labor.¹ Since 100% of respondents reported climbing as part of their regular labor and 99% of respondents reported forerunning, routesetters are clearly engaging in regular climbing activity as part of their labor. Zeidelhack et al.⁹ report that, on average, routesetters spend 30-40% of the day forerunning. This time spent forerunning presents a wide window of exposure to finger injury risk.

As the leading site of injury among rock climbers, fingers represent 30-40% of all rock climbing injuries.^{2,3} Fingers/hands are also the leading injury site among routesetters, albeit at a lesser occurrence (21%). The occurrence of finger injuries in routesetting is more consistent with the rate of finger injuries in construction work (17%).¹ However, the cause of most finger/hand injuries in construction is laceration or fracture, which is inconsistent with the cause of injury reported by routesetters.¹ Thus, routesetters are experiencing a high number of finger/hand injuries likely as a direct result of their climbing/forerunning labor but at an occurrence rate consistent with construction.

Shoulder Injuries

Shoulders accounted for 13.5% of reported routesetting injuries. This is most consistent with the rate of shoulder injuries in climbers (20%),² especially since shoulder injuries account for less than 4% of construction injuries.¹ The leading cause of shoulder injuries was overworking, accounting for 51% of

routesetting shoulder injuries. Too heavy loads was the reported cause of 19% of routesetting shoulder injuries and forerunning was the cause of 13% of the injuries. Since the leading causes of shoulder injuries in construction were strain or contusion, routesetting shoulder injuries are likely derived from climbing associated labor, including forerunning. Thus, shoulder injuries in routesetting represent a rate and causal mechanism most similar to that found in climbing.

Back Injuries

The third highest reported site of injury among routesetters was the back, accounting for 12% of all reported injuries. This injury site is more likely a result of construction labor, where the back is a leading site of injury and accounts for 8.5% of all injuries. Furthermore, with 96% of respondents reporting lifting heavy equipment as part of their regular routesetting labor, and the leading cause of back injuries in routesetters being overworking, this mechanism of injury is consistent with that found in construction labor and would not be found in rock climbing.¹ Additionally, back injuries in construction represent 39% of injuries caused by overworking.¹ Thus, both the occurrence and mechanism of back injury in routesetting is similar to that found among construction workers. Back injuries in routesetting are therefore most likely derived from the constructive aspect of the profession.

Injury Cause Trends

Overworking was the leading cause of injuries in routesetters, accounting for 50-60% of injuries within the top five injury sites. The second leading cause of

injury varied between too heavy loads and forerunning. Overworking suggests a chronic injury that results from cumulative trauma to the injury site. Forerunning and too heavy loads, on the other hand, suggest a more acute mechanism of injury whereby the injury can be traced to a single source event such as an accident. Zeidelhack et al.⁹ reported that 51% of respondents experienced injury caused by “being tired,” while an additional 27% of respondents experienced injury caused by “heavy loads.” Zeidelhack et al.’s findings,⁹ while similar to the mechanisms in this study, also suggest the potential for both chronic injury from cumulative fatigue as well as acute injury from overexertion. Furthermore, these mechanisms are not mutually exclusive: chronic injury may develop from an acute injury and vice versa. Thus, injuries in routesetting occur on both a chronic and acute basis.

Both finger/hand injuries and shoulder injuries are more clearly traced back to the climbing labor involved in routesetting. Back injuries are most likely associated with the constructive aspect of routesetting. However, as has been shown, both construction work and climbing can produce strain on the same injury sites--potentially compounding the potential for injury. Thus, the injury mechanisms in routesetting are a unique product of the scope of labor that requires workers to operate at both a high level of climbing fitness while simultaneously performing regular construction labor.

The causes of injuries in routesetting reflect trends in both rock climbing injury epidemiology and construction injury research. The intersection of these two fields produces a unique injury landscape. While both rock climbing literature

and construction literature provide a partial picture of the injury risk routesetters face, the fields cannot account for the way this intersection of labor will produce unique risks to the routesetting population. Thus, routesetting injury literature must continue to identify and examine the specific impacts of routesetting labor on workers by supplying case-based studies to inform the future of the profession.

2.6.3 Does PPE use affect the likelihood of injury?

The trend of eye injuries in routesetting is similar to that found in construction work. Eye injuries are the fourth leading injury diagnosis among construction workers (accounting for 8% of reported injuries) and the sixth leading injury site among routesetters (accounting for 9% of reported injury sites).¹ Eye injuries among climbers were nonexistent.^{2,3,12} The leading cause of injury among construction workers was an object in the eye and, among routesetters, was a lack of eye protection.¹ These two causes are functionally the same since eye protection serves to prevent debris from entering and harming the eye. Thus, the leading cause of eye injury among both routesetters and construction workers should be interpreted as debris in the eye that may be preventable using adequate PPE.

Tool use trends in routesetting can help explain the rate of eye injury. Although 98% of survey respondents reported using a power tool regularly and 89% reported using an impact tool to install bolts, just 35% of participants reported always using eye protection. These findings are consistent with those by

Zeidelhack et al who reported that 85% of respondents use an impact driver and only 35% use eye protection. Tool use is the most frequently associated activity with eye injury during construction work, and eye injuries are largely preventable using proper eye protection.¹ The OSHA standard for construction work mandates that workers use appropriate eye protection when a hazard exists that cannot be removed through administrative controls.¹ Two-thirds of construction workers who reported an eye injury were not wearing adequate eye protection, and over 80% of routesetters who reported an eye injury were not either.¹ Thus, both construction injury literature as well as OSHA standards suggest that eye injuries are preventable with proper PPE. This fact is reflected in the routesetting injury data, and perhaps confirmed by the routesetting survey conducted by Zeidelhack et al.⁹ who reported that missing or defective PPE accounted for up to 11% of routesetter injuries.

Fortunately, 85% of respondents reported being more likely to use eye protection after experiencing their eye injury(s). This trend of increased PPE use will hopefully increase with the professionalization of routesetting and increased supervision by regulatory bodies such as OSHA.⁷ Additionally, as the culture shifts to promote PPE use, Spangenberg suggests that this will further increase the salience of the issue and increase participation in risk-averse behavior.¹³ Eye injuries are not only expensive to both employers and employees but can have a life-long impact on workers.¹ Industry-wide PPE use, when appropriate, must be considered an essential component of professional routesetting moving forward. If routesetters cannot work due to eye injury, the climbing industry cannot grow.

2.6.4 Does the scope of labor affect the likelihood of injury?

Across all three fields in which scope of labor was analyzed, injury trends emerged across the division of labor. The most linear relationship was found when examining how setting frequency affects injury prevalence, where higher setting frequency was generally linked to a higher injury rate. However, there was one exception to this analysis, where the rate of injury suddenly increased again for the last category of respondents: those who set several times per year. From 5+ days per week down to several times per month, injury rates decreased from almost 86% down to about 42%. Injury rate among those who set several times per year suddenly rose again to about 57%.

Injury Trends Regarding Setting Frequency

First and foremost, the clear relationship between higher setting frequency and injury can be well explained. Higher volume of labor exposes workers to a higher cumulative load on the body, clearly resulting in higher rates of injury. Furthermore, longer cumulative time spend routesetting exposes routesetters to more opportunities for sustaining acute injuries. Thus, those who engage in routesetting labor more frequently will be more susceptible to injury--both chronic and acute.

There are several possible explanations for the sudden increase in injuries among those who set several times per year. First of all, this group had very few respondents (n=7), compared to other groups with more respondents (i.e., n>26). However, the highest frequency group also had only seven respondents so this

explanation would also call into question the broader trend. A second possibility is that those who engage in routesetting labor with such infrequency are not appropriately conditioned to sustain the physical demands from the labor. Thus, these workers are more vulnerable to acute injury caused by inadequate physical conditioning.

The third, and final, possibility is that the workers in this group simply lack the experience or desire to minimize the physical impact from their routesetting labor and are thus more prone to injury. For example, a routesetter working multiple times per week might seek more opportunities to “work smart not hard,” knowing that the cumulative strain from multiple days of setting can be reduced by employing the most efficient setting practices such as not climbing while setting, ascending a rope or ladder only the minimal number of times, and attaching a hold to the wall only once (not constantly rearranging it and wasting energy). The infrequent setter, however, might be unconcerned about such chronic strain since they rarely perform this labor. This infrequent setter might thus make decisions that are less efficient, require more energy, and expose the routesetter to higher physical strain during the given work day--knowing that they will not have to repeat the feat again for some time. The second and third possibilities seem the most likely explanations for the increased rate of injury among those who set several times per year. Further data would be needed to confirm or refute either hypothesis.

Injury Trends Among Those Setting Routes versus Boulders

A relationship also emerged between rate of injury and time spent setting routes versus boulders. The highest rate of injury was found in those who reported setting 100% routes. However, with only two respondents in this group, this data lacks substantial supporting evidence to draw a strong conclusion. Without more data to confirm this finding, the suggested trend may be due to an inadequate number of data points that misrepresents the real trend within the entire population. Furthermore, this finding is not supported by an adequate range of data. While all other groups reported instances of both injury occurrence and lack of injury occurrence, the data in this group only reported injury occurrence.

Among the other groups, however, a potential trend emerged. Data from these groups suggests that specialization reduces the rate of injury. For example, among those who set 100% boulders and those who set mostly routes, the rate of injury was comparably lower than among those who set a more even combination of both routes and boulders. This injury trend may be explained by examining the different labor requirements for setting routes versus boulders. While both those setting routes and those setting boulders are required to transport all their equipment to the worksite at the base of the wall, those setting routes have the added labor of hauling equipment to the top of the wall. This work of maneuvering equipment around the worksite can single-handedly account for vast amounts of work time depending on the efficiency of the routesetter. Those who are well familiar with the proper rope systems may be

able to incur the minimum physical strain to accomplish this task. Meanwhile, those without enough experience or practice may find themselves expending time and energy correcting mistakes, repositioning too frequently, or struggling to maneuver while setting from a rope. For this reason, those setting on ropes frequently may be able to work more efficiently due to their greater experience while those who set from ropes rarely will be more likely to expend excess energy and incur unnecessary physical strain due to their lack of practice.

The theory that those who perform a labor more frequently will develop more efficiency at the given task could explain this injury trend among those who set routes versus boulders. By this logic, those who reported setting all boulders would have a relatively lower injury rate since they do not incur the additional stress from setting on a rope. This theory would also explain why those who spend the majority of their time setting routes would also have a lower injury rate since they would have developed greater efficiency at the tasks simply due to experience.

Injury Trends Among Commercial versus Competition Setters

The trend of injury rate among those setting commercially versus for competition was not as linear as previous injury trends. All respondents to this question reported setting commercially at least 50% of the time, so the data is heavily weighted towards commercial setting with minimal data about competition specific setters. There were three notable correlations in this data. First, those who reported setting mostly commercially (70-90% of the time) and sometimes

for competition (10-30% of the time) reported a much higher injury rate (68-81%) than those who reported setting only commercially (35%). Second, the rate of injury among those who reported setting mostly commercially and sometimes for competition was much higher than the average rate of injury among all routesetters. Third and finally, those who reported equal amounts of time spent setting for competition and commercially reported the lowest injury rate of all the groups (17%).

The Difference between Commercial and Competition Setting

Before examining these relationships, it will be helpful to understand the difference between competition and commercial routesetting. Commercial setting is focused on providing an even distribution of climbs across a range of difficulties on a regular basis. Most routesetters reported setting 2-3 days per week and an average of 5.7 boulders or 2.6 routes. This clearly applies to commercial setting since climbing competitions occur with much less frequency than every week and typically involve a much higher volume of routes. Thus, commercial routesetters are conditioned for setting a relatively low volume of routes multiple times per week.

Competition routesetting exists in contrast to this trend. Competitions are, by definition, designed to challenge competitors, ensure a fair outcome, and produce a winner. Thus, competition setters will typically be setting more complex or difficult routes since they are setting for a more dedicated climber (a competitor). Furthermore, competition setters will be spending more time

ensuring the fairness of a route or a round of climbs as a whole to ensure the fairness of the competition outcome. Finally, competition setters are typically working on a much shorter timeline with perhaps just a few days or up to a full week to set the entire competition. For this reason, competition setters may be setting a much higher volume of routes within a much shorter time, perhaps as many as a hundred boulders over just a few days.

Understanding potential differences between commercial and competition setting poses some interesting inferences about the individual characteristics of the routesetter. Competition routesetters who are working in a much higher volume and at a higher average difficulty of route will need higher climbing fitness to effectively complete their work. Commercial setters, however, might operate at a lower average difficulty and with a much lower volume of routes, necessitating a much lower level of climbing fitness and proficiency compared to the competition setter. While climbers of all levels can theoretically set routes of all difficulties, routesetters must be able to climb near the difficult level they are setting if they must forerun the climb, as 100% of respondents reported doing regularly. Thus, competition routesetters must be stronger climbers with better fitness to meet the higher volume of more difficult climbs required for competition setting.

Variations in Fitness between Commercial and Competition Setters

For the first and second findings, it is clear that the introduction of competition setting in addition to commercial setting increases the injury occurrence among respondents. However, given the low injury occurrence among the group who sets the most for competitions (17%), this trend cannot be due to a higher rate of injury among competition setters. Commercial routesetters demonstrate a low rate of injury, perhaps due to a predictable and consistent volume of labor for which they are regularly conditioned. The introduction of some competition routesetting, however, would create a high strain on the commercial routesetter who might be unaccustomed to the high volume and intensity of competition routesetting. Thus, a commercial routesetter who engages sometimes in competition setting could be more vulnerable to injury due to inadequate fitness, climbing ability, and/or the strain on top of existing commercial setting duties. This explanation would also explain the second finding of a higher than average rate of injury among those who set mostly commercially and sometimes for competition.

The third finding draws on the same process of thinking for the first two findings. Setters who spend an even amount of time setting commercially and for competition will have a better fitness for both since their regular labor includes an equal quantity of both lower volume and lower intensity commercial setting as well as higher volume and higher intensity competition setting. This theory may be supported by a study from Grønhaug et al. who reported that the most experienced climbers in their injury study (those in the international elite group)

reported the lowest rate of injury compared to the intermediate climbers (experienced and elite groups).² The study further suggests that the international elite climbers and the most experienced climbers in the elite group may “have more knowledge of how to train and know their body well enough to train hard without getting injured.”² Thus, if competition setters are composed of stronger climbers, this population may be less susceptible to injury compared to commercial setters, and it would make sense to see this relationship starting at the 50/50 mark for time spent setting commercially/for competition. More extensive data on competition setters would be needed to confirm this relationship.

2.6.5 Limitations

There are several limitations of this study. First, the participant size is smaller than the only published survey on routesetting, *Routesetters Anonymous*, which drew almost 1,000 participants.⁹ This survey drew just under 200 participants in one month. While there are more than enough overall participants to draw conclusions about major questions such as the overall injury rate among routesetters, some specific questions had only a few participants. Questions with less than five answers were not analyzed, and those with a low count of data may not be fully representative of the larger population,⁹ especially given the largely homogenous demographic of responses for this survey.

The overall response rate was likely limited by distribution power. Both this survey and the author’s pilot survey¹⁰ from April 2019 demonstrated a similar response rate having used very similar distribution strategies. The main

difference in distribution methods was the length of time that each survey was open. The pilot survey¹⁰ was open for one week while this study was open for one month. Thus, distribution power is likely a limiting factor in the overall response numbers.

Furthermore, the title of the survey, "Injuries in Routesetting," may have selected a biased sample. Those who have data about their experience with injuries in routesetting may have been more likely to participate, and only those who identify as routesetters might have felt included in the subject population. Routesetters who identify with a different job title (such as builder or course setter) might not feel that the survey applies to them. However, the incentive system used to reward participants could have helped offset this bias by encouraging routesetters of all backgrounds to participate in the survey for the chance to win a prize.

Finally, the distribution method was entirely online and only available for one month. Those without access to the internet would not have had the opportunity to participate. Still others may have been excluded due to the limited time the survey was open if they were otherwise occupied (such as if they were on vacation). Additionally, the survey was primarily distributed within the Facebook group "Routesetters Anonymous" which presents further opportunities for sampling bias. Foremost, the age demographic of those who use Facebook may not fully represent the routesetting population. For example, younger routesetters may prefer other social media platforms while older routesetters may

avoid social media altogether. This could result in a distribution avenue that inadequately captures a representative population of routesetters.

Future surveys might consider a more diverse set of distribution avenues. Such strategies could include recruiting participants in person, via phone, or through advertisements.

2.6.6 Future Directions

The research conducted in this survey has provided a general outline of the routesetting injury landscape. Future research should examine individual components of this landscape for more specific causal relationships between, for example, injury type and mechanism of injury. By exploring specific anecdotal causation of an injury, future research could explore specific preventative methods and novel solutions to the unique risks in routesetting.

It is also important to better understand the specific construction-related impacts on routesetters, particularly through the lens and language of existing literature. Future research should explore the relationship between routesetting injury and ladder use. Ladder fall injuries are a leading cause of injury among construction workers.¹⁴ With 96% (n=137/143) of respondents to this routesetting survey reporting ascending and descending ladders as part of their regular labor, ladders pose significant potential for injury. Additionally, the effects of chronic impact tool usage should be explored as a potential contributing factor of routesetting injury. An industry shift towards hydraulic impact tools and away from conventional impact tools may be occurring. This may be due to the lower

perceived impact of hydraulic impact mechanisms on the body, and a resulting lower risk of chronic injuries such as tendonitis.

Furthermore, it should be noted that this survey was heavily weighted towards commercial routesetting, with most respondents setting commercially at least 50% of the time. More data from competition routesetters will help explain the difference in scope of labor and its resulting impact on injury trends. Finally, future research should examine the relationship between routesetting standards in different countries. With differing regulations and professional standards, not all countries demonstrate the same culture within their professional routesetting programs. Variations to these cultures should be explored and compared to identify specific cultural behaviors that may be increasing or decreasing injury occurrence among routesetters.

2.7 Reference List

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2.8 Appendices

2.8.1 Appendix I: Institutional Review Board Application

Emory & Henry College Exempt Determination Form

ADVISORIES

*Applying for a Grant?
Who can be a Principal Investigator?*

Limitations on Exemptions:

Children: Exemption #2(i) and (j) for research involving survey or interview procedures or observations of public behavior does NOT apply to research in children, except for research involving observations of public behavior when the investigator does not participate in the activities being observed. Exemption #2(ii), where identifiable information is obtained and the IRB conducts a limited IRB review, is NOT applicable to research in children. Exemption #3 does NOT apply to research involving children. [§46.104(b)(3)]

Prisoners: Exemptions do not apply EXCEPT for research aimed at involving a broader subject population that only incidentally includes prisoners. [§46.104(b)(2)]

FDA-regulated research:

Other than exempt category 6, these categories do not apply to research that is also FDA-regulated.

Belmont Principles Apply to Exempt Research:

Even when research is exempt from further requirements of federal regulations, basic ethical standards still apply:

- Except in the case of chart reviews or database research, potential subjects must be provided enough information to be able to choose whether or not to participate. The information would typically include the voluntariness of their participation, the purpose of the research, the nature of the subject's involvement, time commitments, and contact information for the investigator.
- Research data must be handled and stored securely.
- Access to research data must be limited to study team members and other authorized personnel.
- All members of the research team must be current on human subjects training and must have a current conflict of interest disclosure.

If you have any questions about Exemptions, contact the Research Compliance Officer (Bonnie Binkley) or IRB Chair (Dr. Kim Baranovsky): irb@ehc.edu, #6921.

EXEMPTION CATEGORIES

Indicate which category is the best that for your research. More than one category may be applicable.

Note: The research must present no more than **minimal risk** to participants. Minimal risk is defined as: the probability and magnitude of harm or discomfort anticipated in the research are not greater in and of themselves than those ordinarily encountered in daily life during the performance of routine physical or psychological examinations or tests.

The revised common rule contains several categories of research eligible for exemption they are described on a guidance document on the IRB website

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Category	Criteria [45 CFR 46.104(d)]
1	☐ Education research if <u>all</u> of the following conditions are met: ☐ The research will only be conducted in established or commonly-accepted educational settings including but not limited to schools and colleges (may include other sites where educational activities regularly occur); and ☐ The research will involve only normal educational practices, such as (i) research on regular and special education instructional strategies, or (ii) research on the effectiveness of or the comparison among instructional techniques, curricula, or classroom management methods; and ☐ There will be no impact of student's opportunity to learn required educational content or the assessment of educators who provide instruction. (If the research involves significant time and attention away from the delivery of regular curriculum or withholding of standard educational content, this exemption would not apply.)
2	☒ Surveys, Interviews, and Observations <u>if at least one of the first three criteria is met:</u> ☒ (i) Data is anonymous or; ☐ (ii) Data are identifiable but not sensitive; or ☐ (iii) Data are identifiable and sensitive; IRB will that security safeguards are adequate (Adult competent participants only) ☐ If the research involves children as subjects, the research procedures are limited to the use of educational tests (cognitive, diagnostic, aptitude, achievement) or observation of public behavior when the investigator(s) do not participate in the activities being observed
3	☐ Benign Behavioral Interventions <u>if at least one of the first three criteria is met:</u> ☐ (i) Data is anonymous or; ☐ (ii) Data are identifiable but not sensitive; or ☐ (iii) Data are identifiable and sensitive; IRB will that security safeguards are adequate All of the following has to be true for this category: ☐ Only competent adults will be participants ☐ Participants will be informed of the procedures (consent statement) ☐ Procedures are limited to communication or interpersonal contact with the subject: the performance of a cognitive, intellectual, educational or behavioral task; or manipulation of the subject's physical, sensory, social, or emotional environment ☐ The intervention is brief in duration

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	☐ The intervention that is not expected to cause physical or emotional harm, or persistent discomfort; or, generally be experienced by the subject as embarrassing, or generally be offensive. ☐ Data Collection is limited to 1) verbal (oral) or written responses by the subject, 2) data entry by the subject, or 3) observation of the subject, including audiovisual recording ☐ If deception is involved, the potential participant must be informed that he or she will be unaware of or misled regarding the nature or purposes of the research
4	☐ Secondary Research: ☐ (i) information/specimens publicly available or; ☐ (ii) data are recorded without identifiers; or ☐ (iii) Use of identifiable health information covered by HIPAA regulations with either 1) a waiver of HIPAA authorization, 2) covered entity provides de-identified data, 3) covered entity provides a limited data set with a limited DUA in place. (Please note: if the study team interacts with identifiable information, even if identifiable information is not recorded, this does not meet conditions two or three, and a waiver of the HIPAA authorization is required) ☐ (iv) (Very Rare at E&H) Research is conducted by, or on behalf of, a Federal Department or agency using government-generated or government-collected information obtained for nonresearch activities.
5	☐ Research & Demonstration Projects (<u>generally not applicable to research at E&H</u>)
6	☐ Taste and Food Quality and Consumer Acceptance (i) if wholesome foods without additives are consumed, or (ii) If a food is consumed that contains a food ingredient at or below the level and for a use found to be safe, or agricultural chemical or environmental contaminant at or below the level found to be safe, by the Food and Drug Administration or approved by the Environmental Protection Agency or the Food Safety and Inspection Service of the U.S. Department of Agriculture.

STUDY TITLE INVESTIGATOR INFORMATION

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1. Project Title	Routesetting and Injury		
2. Principal Investigator (Last name, First)	Last Name	First Name	
	Beltz, Eleanor		
Must be Faculty or Approved Staff			
Is PI acting as a faculty advisor for this project?	☒ Yes ☐ No		
Type of Research, check all that apply	☒ Undergraduate Honors Thesis ☐ Ampersand Project ☐ First year project ☐ Class Project ☒ Student Driven Research ☐ Faculty Research ☐ Thesis/Dissertation ☐ Research Process Project - SHS ☐ Other: Describe:		
Email	nbeltz@ehc.edu		
Phone	276-944-7245	Department	Athletic Training

3. Co-Investigators and Study Staff : Below, list name, role,, and email for each investigator.

Name	Email Address	Role (e.g., Co-I, Student Co-Investigator, Research Assistant, Student Coordinator)
Henry Dalby	hbdalby16@ehc.edu	Student Co-Investigator

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10. Potential Benefits to subjects

There are no direct benefits to participants.

11. Societal Benefits to subjects/Importance of the Knowledge to be Gained

This research may provide a better understanding of the rate of injury among routesetters. Findings from this survey may impact routesetting, industry healthcare standards.

12. Provisions to Protect Privacy of Subjects and to Maintain Confidentiality of Data

Data will be collected anonymously by Qualtrics. Qualtrics will not record IP addresses from participants. Data exported from Qualtrics will be stored in a password-protected Microsoft Excel document and stored on Microsoft OneDrive. Only the PI and the student PI will have access to these documents and associated passwords.

13. Consent Process

Participants must consent to the survey using the Qualtrics online anonymous data collection consent template. Participants who do not consent will not complete the survey.

14. Expected Outcomes or Results

During the spring 2019 pilot study, we identified that routesetters were getting injured as a direct result from their professional activity. For this expansion on that topic, we aim to better correlate these injuries with their associated causes. We expect these causes may include the following: use of personal protective equipment, changing industry standards, and the scope of labor associated with routesetting.

15. Do you Plan to Share Results with the Community You Studied?

We hope to present and/or publish findings from this survey. Aggregated findings may be shared directly with the route setting community from which we recruited participants.

16. Do you plan to keep any identifiable information for future studies to be conducted by the research team and/or to contact participants about future research projects? Do you plan to use data removed of identifiers for future research?

Not applicable. No identifiable information will be collected.

17. Do you plan to share de-identified data or anonymous data with other researchers not at Emory & Henry?

No.

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18. Additional Information

None.

19. Attachments

Attached to this application is a copy of the survey, including the consent page required to complete the survey.

REQUIRED SUBFORMS

If your project involves the following types of projects, an additional subform will be required:

Secondary Use – HIPAA protected information: Subform M MedRecords ☐ Attached

Audio – Video Recording: Subform H ☐ Attached

FINANCIAL CONSIDERATIONS

Financial Interest in Research

Do any members of the research team or any members of their immediate families have any financial interest in the sponsor of this research and/or in the results of this research? ☐ Yes ☒ No If Yes, complete IRB Form 3.

Project Funding

Is this project supported by any source of funding, either internal (such as a faculty research award) or by an external grant or contract? ☐ Yes ☒ No If Yes, complete IRB Form 4.

Please send an electronic Word attachment (not scanned) of this application and any accompanying materials (recruitment, consent form, data collection instruments such as surveys, or interview questions) as separate files to irb@ehc.edu

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Thank you for taking your time to promote ethical human participant research at Emory & Henry College!

Assurance of Principal Investigator

Complete this Section if Students Will Not Function Independently in Conducting this Research.

Electronic signature. My checking of the box below is representation of my signature and is legally valid and binding as if I had signed the document with ink on paper in accordance with Virginia's Uniform Electronic Transactions Act (UETA) and the Electronic Signatures in Global and National Commerce Act (E-SIGN) of 2000.

As the Principal Investigator, I assure that:

- I will conduct this study as described in this application and accompanying documents.
- I will report immediately to the E&H Institutional Review Board any changes in the procedure; injury to a research participant; or any problems that involve risk or the possibility of risk to participants or others.

X
NAME

Eleanor M. Beltz

Date:

12/1
3/2019

Email:

nbeltz@ehc.edu

Complete this section if a Student or Students Will Have a Major Role in Conducting this Research.

Electronic signature. My checking of the box below is representation of my signature and is legally valid and binding as if I had signed the document with ink on paper in accordance with Virginia's Uniform Electronic Transactions Act (UETA) and the Electronic Signatures in Global and National Commerce Act (E-SIGN) of 2000.

As the Principal Investigator and Faculty Supervisor for the research proposed in this application, I assure that:

- The student(s) is/are sufficiently knowledgeable about the regulations and policies governing research with human subject, and has/have sufficient training and experience, to conduct this particular study in accord with the approved protocol.
- I have reviewed the application and all associated materials;
- I will meet with the student investigator(s) as necessary to monitor study progress;
- I will assist and advise the student(s) in reporting immediately to the E&H Institutional Review Board any changes in the procedure; injury to a research participant; or any problems that involve risk or the possibility of risk to participants or others.

X
NAME

Eleanor M. Beltz

Date:

12/13
/2019

Email:

nbeltz@ehc.edu

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INSTRUCTIONS AND NOTES

Applying for a Grant? Certain private and federal granting agencies are reluctant to accept a determination of exemption as evidence of adequate review. If you are applying for a grant, please ask the agency whether exemption is acceptable. If not, you will have to submit a regular IRB application. [Return to first page](#)

Principal Investigator A principal investigator must be a faculty member, qualified by training and experience to assume responsibility for the conduct of the research. A student, trainee, or visiting researcher cannot be a principal investigator. [Return to first page](#)

Objective, Aim or Purpose Briefly state your research question or hypothesis

Background Briefly describe any relevant background, any literature searches performed and where this fits in the body of the literature, and anticipated plans for disseminating results.

Methods Explain how the study will be conducted from start to finish. Be specific about the methods, instrumentation, and types of data to be collected. Include the name and name and description of all surveys and instruments that will be used. If a survey or instrument has been developed by the investigator, this should be stated. **Note regarding "anonymous" internet surveys:** If you intend to collect data anonymously, you must use Qualtrics, and you must indicate you will turn off collecting IP addresses.

Location If the proposed research will be conducted off the Emory & Henry campus, please include permission from an authorized official of the site.

Subject Population Describe the subject population, age range of participants, and whether or not participants will be students or employees of the institutions/organizations from which you intend to recruit

Recruitment of Research Subjects Describe how you will recruit research subjects and how many you anticipate recruit

Risks Risks are usually classified as physical, psychological, legal, financial, and loss of personally identifiable confidential information. Please avoid saying there are no risks associated with the study. Instead, you may state that the risks are no greater than those encountered in everyday life. **For Design Behavioral interventions.** Ordinary, mild, transient forms of discomfort, such as the stress associated with completing a timed cognitive task, anxiety about performance, and boredom, are consistent with the intent of the exemption. Similarly, while research cannot meaningfully eliminate all risk of embarrassment or offense, the research should include only interventions that the researcher has no reason to think subjects will find offensive or embarrassing considering the characteristics of the subject population, the research context, and how they might impact the subject's experience of the research intervention.

Special Protections Special safeguards are usually required when vulnerable persons (children, pregnant women, persons with diminished mental capacity, economically or educationally disadvantaged) are involved in research studies. Use Form A for Children; Form C for Pregnant Women; Form for Persons with Impaired Decision-Making Capacity.

Payment to Research Subjects If you will pay the participants anything (cash or course credit), specify the amount and whether it will be a single payment or distributed over a number of sessions; describe how payment will be made if you have assured participants of anonymity; and indicate the source of funding.

Potential Benefits to Subjects Describe the potential benefits to individuals participating in the study. In some studies, there are no direct benefits to the individuals participating in the study. If this is the case this should be stated.

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Request for Modification of Approved Human Participant Research

Emory & Henry College

Instructions: Download this form as a word document (do not convert to a Google Doc) and complete it. Send the request form electronically to: IRB@ehc.edu. Note: Checkboxes can be checked by putting an "x" in the box and text should be placed inside of the fillable boxes labeled "Click or tap here to enter text."

IRB Modification Request Form

Section 1: Background Information	
Principal Investigator (last name, first)	Bello, Nora
Project Title	Inquiries in Redempting
IRB Protocol #	Click or tap here to enter text.
Research Type (check one)	☒ Expedited ☐ Full Review
Email	nrbello@ehc.edu
Phone	276/944-7243
Campus Office Address	Department: Academic Training ☐ Main ☒ Health Science
Co-Investigator/Other Contact (if applicable)	Duffy, Henry
Email	hduffy1@ehc.edu
Phone	637/599-4251
Funding Agency (if applicable)	Click or tap here to enter text.

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Mod Request January 12, 2018

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Societal Benefits/Importance of the Knowledge to be Gained Describe the potential benefits to society or value of the knowledge to be gained.

Provisions to Protect Privacy of Subjects and to Maintain Confidentiality of Data? Strongly recommended for all applications. Required for Exemptions 2(iii), 3(iii), 7, and 8. This can include collecting anonymous data, use of encrypted files, locked file cabinets with identifying information stored separately from the data, and use of OneDrive. Check with E&H IT if you need additional assistance for storage of electronic data.

Consent Process Except in the case of chart reviews or database research, potential subjects must be provided enough information to be able to choose whether or not to participate. Depending on your research, you should either use the online anonymous data collection consent template or the social behavioral – exempt template to create a consent statement. These statements will not typically require a signature, though in some cases the IRB may require it. In this section, describe how the consent process will be conducted.

Sharing Results Please do not describe plans to publish in professional journals or present at professional meetings.

Additional Information Use this space for any information not captured in the preceding sections that might be useful to the Reviewer.

Attachments List any material that will be shown to participants such as information sheets, consent forms, and final versions of surveys. Upload the Attachments as separate documents

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IRB Version date 4-4-2019

Name	Email Address	Role
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Please also Note: All members of the study team are required to complete CITI human subjects' Protections training. The current requirements are listed on the Emory & Henry Research Compliance website Human Subject Protection Training <https://www.ehc.edu/research-compliance/human-subject-protection-training/>. Training will be confirmed by the IRB office, and IRB approval for the study cannot be issued until all study team members have satisfied the requirements.

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Section 2: Modifications Overview		
1. Check all that apply:		
☐ Adding key personnel or research assistants	☐ Removing key personnel or research assistants	
☐ Location changes	☒ Adding funding source → include copy of the grant	
☐ Conflict of interest changes	☐ Removing a funding source	
☐ Inclusion criteria changes	☐ Increasing participants	
☐ Exclusion criteria changes	☐ Decreasing participants	
☐ Data collection method and/or new advertisement tools	☐ Study procedures	
☒ Compensation	☐ Changes to or addition of informed consent/assent/parental permission form	
☐ Title change	☐ Instruments: adding, removing, changing	
☐ Reopening enrollment	☐ Risks	
☐ Other		
Section 3: Personnel		
1. If adding key personnel (involved in recruitment and/or data collection) list the following information for each person.		
☐ N/A		
Name	Email Address	Role (e.g., Co-1, Student Co-Investigator, Research Assistant, Student Assistant, Research Coordinator)

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3. If no other modifications are being requested, skip to Section 6.

Section 4: Current Enrollment Status

1. If increasing, decreasing, or reopening enrollment, please indicate: reopening enrollment

☐ N/A

Total number of subjects currently **APPROVED** for enrollment

117

Total number of subjects currently **ENROLLED** in the study

117

Currently Approved **AGE RANGE** of participants

18 years or older

Section 5: Serious Modification Reasons (Formal Risk-Related Concerns)

1. Briefly describe the anticipated modification(s) and reasons for the request. (Attach the revised IRB application any revised instruments, recruitment materials, updated consent form, etc.)

The incentives for participants are being changed after receiving funding approval from the Dr. Holmwood and the Human Experiment. Participants will now be eligible to enter a raffle to win one of 5,000 gift certificates.

2. Does the modification increase or decrease the level of risk for of participants?

☒ No ☐ Yes

If the modification **increases the level of risk**, please explain and include what measures will be taken to minimize these additional risks.

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Mod Request September 7, 2018

2. If removing key personnel or research assistants, list their names and titles (if applicable) below and explain the reason for their removal

☒ N/A

Name	Role (i.e., Co-I, Student Co-Investigator, Research Assistant, Student Assistant, etc.)	Reason for Removal
Name	Blank	Click or drag here to enter text.
Name	Blank	Click or drag here to enter text.
Name	Blank	Click or drag here to enter text.
Name	Blank	Click or drag here to enter text.
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Section 6: INVESTIGATOR STATEMENT OF COMPLIANCE

Electronic signature. My checking of the box below is representation of my signature and is legally valid and binding as if I had signed this document in ink. I have read and understand the terms of the Human Experimentation and the Electronic Signature and Consent Act (HESCA) and the Electronic Signature and Consent Act (ESCA) and the Electronic Signature and Consent Act (ESCA) and the Electronic Signature and Consent Act (ESCA).

As the **Principal Investigator**, I certify that:

- All information provided is accurate and that procedures involved in this project were conducted in strict accordance with federal regulations and Emory & Henry policies governing human subject research.

☐

Name: **Eleonor Baite**

Date:

12/13/20

Email: **mbaite@ehc.edu**

Please send an electronic word or pdf attachment (not Google doc) of this application and any accompanying materials (abstracts, publications, etc.) as separate files to IRB@ehc.edu.

Thank you for taking your time to promote ethical human participant research at Emory & Henry College!

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N/A

3. Does this modification(s) change the benefits to participants, and/or society (contribution to generalizable knowledge)?

☐ No ☒ Yes

If yes, explain how.

At the end of the survey, participants will be asked if they would like to be entered in a raffle to win one of 5 gift card incentives for completing the survey. The raffle will be conducted by Emory & Henry College. The raffle will be conducted by compiling the list of names for those that provided them and randomly picking three winners via an online randomized raffle application.

4. May this modification affect current participants' willingness to participate in the study (i.e., revised study procedures, change in compensation, etc.)?

☐ No ☒ Yes

If yes, explain how.

Participants will have a greater incentive to participate for the chance to win prize.

5. If new information may affect currently enrolled participants, how will they be informed about the modification changes, indicate below.

☒ Participants will complete a new informed consent/assent/permission form. -- Submit the new informed consent form for review.

☐ Participants will complete an addendum informed consent/assent/permission form. -- Submit the addendum form for review.

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Mod Request September 7, 2018



INSTITUTIONAL REVIEW BOARD
Room 008 Wiley Hall
Emory, VA
Email: IRB@EHC.EDU

Eleanor (Nora) Beltz, PhD, ATC, CSCS, NASM-CES
Assistant Professor | Department of Athletic Training
School of Health Sciences | Emory & Henry College

From: IRB Administrator/IRB Member, Bonnie Binkley

Date: 12-18-2019

RE: Modification and Continued Exempt Determination by the IRB under 45 CFR 46.104(d)(2)(i)

Study #: 0028_19

Study Title: Routesetting and Injury

The Administrative Section of the Emory & Henry Institutional Review Board (IRB) reviewed your application for revision of your project previously determined to be exempt.

The IRB determined that your revision application is eligible for expedited review under 45 CFR 46.110(b)(2), and that your study remains eligible for exempt status. The modifications to the study as submitted are acceptable, and the study continues to be exempt under 45 CFR 45 CFR 46.104(d)(2)(i). **Any new modifications to the study must be reviewed** by the IRB. The IRB will, with each modification, determine whether the study continues to be exempt.

Current Modifications Include:

1. Change in incentive to enter a raffle to win a \$100 gift certificate to an outdoor retailer of the participants choice, with 5 winners selected.
2. Consent statement and recruitment material updated to reflect change in incentive.
3. Drs. Boltwood and Davis were removed as investigators as they are not involved in data collection or interaction with the data.

Exempt Determination Conditions:

Emory & Henry College Policies: All individuals engaged in research with human participants are responsible for compliance with the College policies and procedures, and IRB determinations.

Post Office Box 947 • Emory, Virginia 24327-0947
256.944.4121 • www.ehc.edu

Principal Investigator Responsibilities: As Principal Investigator, you are responsible for ensuring compliance with all applicable FERPA regulations, Emory & Henry College policies, state and federal regulations. Every effort should be made to ensure that the project is conducted in a manner consistent with the three fundamental principles identified in the Belmont Report: respect for persons; beneficence; and justice. The Principal Investigator is responsible for ensuring that all study personnel have completed the online human subjects training requirement.

Modifications and Addendums: IRB approval must be sought and obtained for any proposed modification or addendum (e.g., a change in procedure, recruitment, personnel, study location, consent, study instruments) to the IRB approved protocol, and informed consent statement/process before changes may be implemented, unless changes are necessary to eliminate apparent immediate hazards to participants. Changes to eliminate apparent immediate hazards must be reported promptly to the IRB.

Prompt Reporting of Events: Unanticipated Problems involving risks to participants or others; serious or continuing noncompliance with IRB requirements and determinations; and suspension or termination of IRB approval by an external entity, must be promptly reported to the IRB.

Closing a study: Although submission of continuing review is not required, when research procedures with human subjects are completed, please complete the Request for Closure of IRB review form and send it to irb@ehc.edu. This helps us to keep track of studies that are currently open on campus.

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256.944.4121 • www.ehc.edu

2.8.2 Appendix II: Survey Questions

Dalby Honors Thesis - Injuries in Routesetting

Start of Survey

Q1

Study Title: Injuries in Routesetting

Principal Investigator(s): Nora Beltz, PhD, ATC, CSCS, NASM-CES, Athletic Training Department at Emory & Henry College School of Health Sciences

Co-Investigator(s): Henry Dalby, Geography Department undergraduate student at Emory & Henry College

Supported By: This research is supported by Emory & Henry College.

What is the purpose of this study?
This study involves survey research regarding the relationship between routesetting and injuries related to this profession.

Why am I being asked to take part in this research study? You are being asked to participate because you regularly engage in routesetting activities. I am asking you to take part in this research because we are interested in understanding the relationship between routesetting and injury.

What should I know about a research study? Whether or not you take part is up to you. You can choose not to take part. You can agree to take part and later change your mind. Your decision will not be held against you. **What I will do if I participate in this study?** You will be asked questions regarding your participation in routesetting activities and your experience with injury related

to routesetting.

How long will it take me to do this? The study should take you under 10 minutes to complete.

Is there any way being in this study could be bad for me? This research poses no risks beyond those encountered in daily life.

What are the potential benefits of participating in the research? This research will provide a better understanding of the proliferation of work-related injuries in the professional field of routesetting. Upon completion of the survey, you will be invited to enter into a raffle to win a \$100 gift certificate to an outdoor retailer of your choice. There will be 5 winners selected.

Will anyone know what I do or say in this research (Confidentiality)? No identifiable information will be collected and no attempt will be made to identify the participant. We will not collect IP addresses.

What happens if I do not want to be in this research or if I say “Yes”, but I change my mind later? Participation in research is voluntary. You can decide to participate or not to participate. You can decide not to participate in this research or you can start and then decide to leave the research at any time and it will not be held against you. To do so, simply exit the survey.

Who can I talk to if I have questions about the study or if I feel hurt by the

research study?

If you would like to contact the Student Investigator in the study to discuss this research or if you feel hurt by the research study, please e-mail Henry Dalby at hbdalby16@ehc.edu. If you would like to contact the Principal Investigator in the study to discuss this research or if you feel hurt by the research study, please e-mail Dr. Nora Beltz at nbeltz@ehc.edu.

How can I contact someone outside the research team for information about this research? If you wish to talk to someone other than the researcher(s) about: questions about your rights as a research participant; wish to obtain information; report a case of research-related injury; ask questions, discuss any concerns about this research; or wish to offer input about this research, please contact (anonymously if you wish): Emory & Henry IRB
P.O. Box 947
Emory, Va. 24327
irb@ehc.edu (276)944-6921
Statement of Consent to Participate.
By continuing with this survey I am indicating that I am 18 years of age or older and that I consent to participate in this research.

- ☐ I consent. Begin the study.
 - ☐ I do not consent. I do not wish to participate.
-

B1 Did you respond to the "Injuries in Routesetting" survey from April 2019?

- ☐ Yes
 - ☐ No
-

B2 Thank you for your contribution to our pilot study. Thanks to your participation in the spring, we are able to launch this improved questionnaire to better understand the injuries in routesetting. While some of the original questions from the pilot have been included below, we highly appreciate your updated response.

Thank you for your participation in the previous study, and your continued interest in supporting the professional development of the field of routesetting.

B3 How old are you?

▼ Under 18 ... 85 or older

B4 What is your gender?

- ☐ Male
 - ☐ Female
 - ☐ Non-binary
 - ☐ Undisclosed
-

B5 What is your ethnicity?

- ☐ White
- ☐ Black or African American
- ☐ American Indian or Alaska Native
- ☐ Asian
- ☐ Native Hawaiian or Pacific Islander
- ☐ Other

B5.5 What is the highest degree or level of education you have completed?

- ☐ Less than high school
- ☐ High school graduate
- ☐ Some college
- ☐ 2 year degree
- ☐ 4 year degree
- ☐ Professional degree
- ☐ Doctorate
- ☐ Other

B6 How many years have you been climbing?

▼ 1 ... 50+

B7 On average, how many hours do you rock climb per week (outside of routesetting)?

▼ 0 ... 40+

B8 What is your physical activity level outside of rock climbing/routesetting?

- ☐ Sedentary (less than 30 min exercise/day)
- ☐ Moderate (30-60 min exercise/day)
- ☐ High (over 60 min exercise/day)

B9 On average, how many complete days off from physical activity do you have per week?

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7

C1 How many years have you been routesetting?

▼ 1 ... 50

C2 How many gyms/companies do you set for?

▼ 1 ... 10+

C3 Where do you currently routeset?
(check all that apply)

- ☐ Home gym
- ☐ Recreation center
- ☐ School
- ☐ Commercial gym
- ☐ Other

C4 Where is the gym/company that you
do the majority of your routesetting?

- ☐ City

- ☐ Country

- ☐ Zip

C5 Are you employed by a climbing
facility or self-employed?

- ☐ Employee
- ☐ Self-employed
- ☐ Other

C6 Do you think you will be routesetting
in 5 years?

- ☐ Yes
- ☐ No
- ☐ Maybe

C6.5 Are you paid to routeset?

- ☐ Yes
- ☐ No

C7 How are you compensated for
routesetting?

- ☐ Per route
- ☐ Hourly
- ☐ Salaried
- ☐ Other (please describe)

C8 Is routesetting your primary source
of income?

- ☐ Yes
- ☐ No

C9 On average, how often do you set?

- ☐ More than 5 days per week
- ☐ 3-4 days per week
- ☐ 2-3 days per week
- ☐ 1-2 days per week
- ☐ Several times per month
- ☐ Several times per year
- ☐ Less than once per year

C10 Do you set routes, boulders, or both?

- ☐ 100% routes - 0% boulders
- ☐ 90% routes - 10% boulders
- ☐ 80% routes - 20% boulders
- ☐ 70% routes - 30% boulders
- ☐ 60% routes - 40% boulders
- ☐ 50% routes - 50% boulders
- ☐ 40% routes - 60% boulders
- ☐ 30% routes - 70% boulders
- ☐ 20% routes - 80% boulders
- ☐ 10% routes - 90% boulders
- ☐ 0% routes - 100% boulders

C11 Do you mostly set commercially or for competitions?

- ☐ 100% commercial - 0% competition
- ☐ 90% commercial - 10% competition
- ☐ 80% commercial - 20% competition
- ☐ 70% commercial - 30% competition
- ☐ 60% commercial - 40% competition
- ☐ 50% commercial - 50% competition
- ☐ 40% commercial - 60% competition
- ☐ 30% commercial - 70% competition
- ☐ 20% commercial - 80% competition
- ☐ 10% commercial - 90% competition
- ☐ 0% commercial - 100% competition

C13 How many boulder problems do you set on average in one day?

- ☐ Amount of boulders/day

C13.5 What is the average wall height for bouldering?

- ☐ Wall height (meters)

C14 How many routes do you set on average in one day?

- ☐ Number of routes/day

C14.5 What is the average wall height for routes?

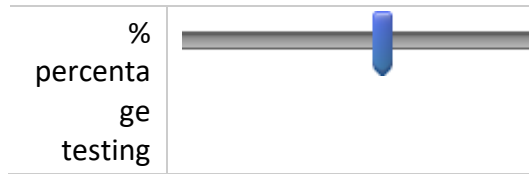
- ☐ Wall height (meters)

C16 What activities apply to your work? (check all that apply)

- ☐ Hold stripping
- ☐ Hold cleaning
- ☐ Hold sorting
- ☐ Route setting
- ☐ Testing/forerunning
- ☐ Other (please describe)

C17 What percentage of your boulder setting time is spent testing/forerunning?

0 100



C17.5 What percentage of your route (roped climbing) setting time is spent testing/forerunning?

0 100



C18 As part of your routesetting activities, do you do any of the following: (check all that apply)

- ☐ Ascend/descend ladders
- ☐ Operate power tools
- ☐ Lift heavy boxes/equipment
- ☐ Climb
- ☐ Belay
- ☐ Ascend a fixed rope
- ☐ Haul heavy equipment up a rope
- ☐ Operate a lift

C19 Which tool do you primarily use for bolts?

- ☐ Impact
- ☐ Non impact
- ☐ By hand

C20 Is your impact tool a conventional impact mechanism or an oil impulse/hydraulic tool?

- ☐ Conventional impact
- ☐ Oil impulse/hydraulic
- ☐ Unknown
- ☐ Other

D0 Do you have health insurance?

- ☐ Yes
- ☐ No

D0.5 Is your health insurance provided through a climbing facility or a different job?

- ☐ Provided through a climbing facility
- ☐ Provided through a different job
- ☐ Provided independently
- ☐ Other

D0.75 Do you warm up prior to performing routesetting labor?

- ☐ Yes
- ☐ No

D1 How often do you use protective eye-wear? (safety glasses)

- ☐ Always
 - ☐ Often
 - ☐ About half the time
 - ☐ Rarely
 - ☐ Never
-

D2 How often do you use ear protection?

- ☐ Always
 - ☐ Often
 - ☐ About half the time
 - ☐ Rarely
 - ☐ Never
-

D2.5 How often do you wear a helmet when setting routes (not boulders)?

- ☐ Always
 - ☐ Often
 - ☐ About half the time
 - ☐ Rarely
 - ☐ Never
-

D3 Are there other forms of personal protective equipment (PPE) that you frequently use?

- ☐ No
 - ☐ Yes (please describe)
-
-

D3.5 How do you feel about the level of safety in your workplace environment?

- ☐ Very high
 - ☐ High
 - ☐ Neutral
 - ☐ Low
 - ☐ Very low
-

D26.5 Have you seen any industry shifts regarding the health, wellness, and safety of routesetters?

- ☐ Yes
 - ☐ No
 - ☐ Other (please describe)
-
-

D26.75 Please describe the industry shifts you have seen regarding the health, wellness, and safety of routesetters:

D4 Have you increased frequency of personal protective equipment (PPE) use due to any of the following? (select all that apply)

- ☐ OSHA/regulatory body standards
 - ☐ Workplace policy
 - ☐ Previous injury
 - ☐ Other
-
-

D5 Have you ever experienced an injury related to routesetting?

- ☐ Yes
- ☐ No

D6 In your career, have you experienced an injury to any of the following: (choose all that apply)

- ☐ Back
- ☐ Shoulder
- ☐ Finger/Hand
- ☐ Knee
- ☐ Wrist
- ☐ Elbow
- ☐ Foot/Ankle
- ☐ Eye
- ☐ Ear
- ☐ Head/face
- ☐ Torso
- ☐ Pelvis/Hip
- ☐ Other (please describe)

D7 Please estimate the number of injuries you have sustained over your entire routesetting career?

▼ 0 ... 100+

D8 Please estimate the number of injuries you have sustained over the past year of routesetting?

▼ 0 ... 100+

D8.5 Are you more likely to use Personal Protective Equipment after your injury(s)?

- ☐ Yes
- ☐ No

D9 What was the cause of your back injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D10 What was the cause of your shoulder injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D10 What was the cause of your finger/hand injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D11 What was the cause of your knee injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D12 What was the cause of your wrist injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D13 What was the cause of your elbow injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D14 What was the cause of your foot/ankle injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D15 What was the cause of your torso injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D16 What was the cause of your
pelvis/hip injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D17 What was the cause of your "other"
injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D18 What was the cause of your eye
injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D19 Were you using protective eye-
wear when you sustained an eye injury?

- ☐ Yes
- ☐ No
- ☐ Varied (multiple eye injuries)

D20 What was the cause of your ear
injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D21 Were you using ear protection
when you sustained an ear injury?

- ☐ Yes
- ☐ No
- ☐ Varied (multiple ear injuries)

D22 What was the cause of your
face/head injury? (select all that apply)

- ☐ Missing or defective PPE
- ☐ Missing or inadequate safety infrastructure
- ☐ Work area not clear or closed off
- ☐ Too heavy loads
- ☐ Overworked
- ☐ Other (please describe)

D23 Were you wearing a helmet when you sustained a face/head injury?

- ☐ Yes
- ☐ No
- ☐ Varied (multiple face/head injuries)

D24 Has your injury or injuries affected your ability to perform routesetting labor?

- ☐ Yes
- ☐ Sometimes
- ☐ No

D25 Does your job provide free or subsidized access to any of the following healthcare:

- ☐ Physiotherapy
- ☐ Yoga
- ☐ Regenerative training
- ☐ No, no free/subsidized health services
- ☐ Other (please describe):

D26 When you get injured, do you seek medical care or rehabilitation? (check all that apply)

- ☐ Yes, formal (from a licensed medical practitioner)
- ☐ Yes, informal (from yourself, family, friends, coworkers etc.)
- ☐ No
- ☐ Other (please describe)

D27 Is there anything else pertaining to the health and wellness of routesetters that you would like us to know?

Q81 Would you like to enter a raffle for the chance to win a prize?

- ☐ Yes
- ☐ No

End of Survey

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